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Stated or Sampled? Pass@k, Voting and Agreement Misjudge the Differential Diagnoses of Post-Trained LLMs Answering Directly

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Abstract

A language model's differential diagnosis can be read from the list it states in one answer or from the spread of its answers over repeated samples, the readout behind pass@k, self-consistency and sampling-based uncertainty. We compare the two readouts across base and post-trained versions of open 4–72B models from four families (Qwen, OLMo, Mistral and the medical MedGemma) on DDXPlus, whose synthetic patients carry a reference differential, and on 897 published case reports. Post-training collapses the samples of models answering directly: post-trained models give 1.3–2.4 distinct answers per 32 samples at temperature 0.7, against 3.0–4.9, and with each model at its best prompt they lose 12–30 points of pass@32 (6–18 after correcting for chance). One stated list of five does not collapse, so the readouts reverse: for every base model, 32 samples name the true diagnosis more often than one list (by 4.5–28.6 points), whereas one post-trained list names it more often than 32 samples (by 2.1–19.6 points; significant in seven of eight pairs, in three after correcting for chance), and zero-shot, for 7–20% of patients the list names it while none of the samples does. Zero-shot, one list also contains the true diagnosis more often than the base model's for every Qwen model, and on the case reports it covers as many of the alternatives the authors considered as before (equivalent within 5 points), while the samples cover 8–14 points fewer for four of five pairs; for the reports' final diagnoses, however, 32 post-trained samples still beat one list in four of five pairs. Tools that read the differential from samples lose value: majority voting barely helps, and 19–61% of post-trained models' wrong answers are repeated by every other sample, so agreement cannot flag them (1–9% for base models). Much of the first-answer gain is prompt format: once base models get the same worked examples and both decode greedily, it is no longer significant in four of six pairs. Studies of diagnostic breadth should report which readout they use, and systems that need several candidates should read them from one stated list, or the union of a few.

Data and Code Availability. All data are public: DDXPlus (Fansi Tchango et al., 2022b) (synthetic patients), Phenopacket Store v0.1.27 (Danis et al., 2025) and MedCaseReasoning (Wu et al., 2025b) (cases from published literature). All models are public checkpoints (Table 6). Prompts and an example vignette are in Appendix B; code and sampled outputs will be released with the camera-ready version.

Institutional Review Board (IRB). This work uses only public synthetic or previously published, de-identified data and involves no human subjects, so it did not require IRB review.

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1. Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. A language model's differential can be read in two ways: from the list it states in one answer, or from the distribution of its answers over repeated samples. Much of the general evaluation and uncertainty toolkit uses the second. Pass@k, self-consistency voting (Wang et al., 2023) and sampling-based uncertainty (Kuhn et al., 2023; Farquhar et al., 2024) draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples (Yue et al., 2025; Wu et al., 2025a; Oh et al., 2026). The two readouts agree for a model that samples from the distribution it states. Instruction-tuned models do not: they verbalize distributions they fail to sample from (Meister et al., 2025; Zhang et al., 2026; Jang et al., 2026), and one aligned answer can cover what many base-model samples spread across calls (Lake et al., 2025). We ask what this means for differential diagnosis, where DDXPlus cases have a reference differential and every case a true diagnosis, and what it does to the tools that read the differential from samples. This is a replication and extension in a clinical setting, not a new phenomenon. Our central claim is that for post-trained models answering directly, the differential should be read from the stated list rather than from samples: one list keeps its coverage, while the samples, and the tools built on them, lose breadth and signal. We call a readout *narrower* when it covers fewer of the conditions the reference warrants; a single list can keep its coverage while lists vary less from call to call, so we report breadth within one list and across calls separately.

The design holds the pretrained weights fixed and varies only the post-training: base and instruct pairs for Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B, and Qwen2.5-72B and the medical MedGemma-4B as later checks; every released stage (SFT, DPO, RL) of OLMo 3 7B (Team Olmo et al., 2025) and OLMo 3.1 32B, plus a 7B checkpoint trained with RL directly from the base; three medical models fine-tuned from Qwen2.5-7B-Instruct; and Qwen3-8B with its thinking mode on. Each final pair is also run with worked examples in the prompt, so that base models answer in the requested format. Table 1 lists our findings with their main evidence and where they weaken; Appendix A gives the full evidence and checks for each.

2. Related work

Post-training and the sampled distribution. RL with verifiable rewards (RLVR) improves pass@1 while base models win at large k on math and code (Yue et al., 2025; Wu et al., 2025a); a "sharpening tax" appears across 14 base/post-trained pairs on agentic tasks (Oh et al., 2026), though RL can expand the boundary in domains under-represented in pretraining or with prolonged training (Cheng et al., 2025; Liu et al., 2025b), and policy entropy collapses early in RL training (Cui et al., 2025). Instruction tuning and RLHF reduce output diversity (Kirk et al., 2024; West and Potts, 2025; Jiang et al., 2025; Murthy et al., 2025), more so with scale for SFT (Springer et al., 2026), and concentrate next-token distributions so that outputs barely respond to decoding (Yang et al., 2026). Where diversity collapses depends on the pipeline: at DPO in OLMo and OLMo 2 (Peeperkorn et al., 2025) and in a stage ablation by Liu (2026), at each OLMo-2 stage for stated-versus-

Finding	Main evidence	Weaker or fails
Sampled answers collapse (§4.1)	1.3–2.4 distinct answers per 32 samples, against 3.0–4.9; pass@32 −12 to −30 points at each model's best prompt (−6 to −18 after chance correction)	chance-corrected loss n.s. for Qwen3-8B, Mistral and MedGemma
The readouts reverse (§4.2)	base models: 32 samples beat one list by 4.5–28.6 points; post-trained: one list beats 32 samples by 2.1–19.6 (seven of eight significant); shift 17–41 in every pair	a consequence of the collapse (equal-size readouts within 5.2 points); chance-corrected lead significant in three of eight
One list keeps its coverage (§4.2)	recall@5 +12 to +16 points zero-shot for every Qwen model and OLMo 3.1 32B; case reports: authors' alternatives covered as before (equivalent within 5 points)	not OLMo 3 7B; weaker with worked examples at 32B; five averaged base-model lists match one post-trained list
Tools built on samples lose value (§4.3)	voting gains at most 1.9 points after post-training; agreement AUROC falls in every pair; 19–61% of post-trained errors are repeated by every sample	the stated probability flags only 5–29% of those silent errors
Case reports (§4.3)	post-trained samples cover 8–14 points fewer of the authors' alternatives (four of five pairs)	for the final diagnosis, samples beat one list in four of five pairs

Table 1: Findings at a glance (DDXPlus unless stated; differences are post-trained minus base unless stated; eight pairs include Mistral-7B and MedGemma-4B with worked examples).

sampled distributions (Jang et al., 2026), and set by training data across OLMo 3's Think, Instruct and RL-Zero lineages (Karouzos et al., 2026). Base models given a few in-context examples approach aligned models (Lin et al., 2024), and theory and experiments cast post-training as sharpening the base model's own distribution toward its mode (Huang et al., 2025; Agarwal et al., 2025; Karan and Du, 2026), or as distilling its majority vote (Prasad et al., 2025). Large-k pass@k over few answer options is inflated by chance (Dragoi et al., 2025).

Stated versus sampled distributions and uncertainty. Models asked to verbalize a distribution match it better than their samples do (Meister et al., 2025; Jang et al., 2026), and they struggle to draw from target distributions directly (Gu et al., 2026; Zhao et al., 2026; Zhang et al., 2024); base models' token probabilities are better calibrated than their post-trained versions' (Kadavath et al., 2022). Verbalized sampling keeps most of the base model's diversity through Tülu 3's checkpoints (Zhang et al., 2026), and aligned answers aggregate content that base models spread across samples on open-ended questions with conflicting perspectives (Lake et al., 2025). Our stated readout is the verbalized top-k elicitation of Tian et al. (2023); asking for several answers in one call gives more diversity at equal quality than independent sampling (Troshin et al., 2025). For error detection, sample consistency usually beats verbalized confidence and hybrids beat both (Xiong et al., 2024; Lyu et al., 2025), including for diagnosis (Savage et al., 2025; Ren et al., 2026), confidence-weighted voting improves self-consistency (Taubenfeld et al., 2025), and agreement cannot flag errors a model repeats consistently (Tan et al., 2025); aligned models' samples can collapse enough to leave sampling-based uncertainty without signal (Liu, 2026), and cross-

model disagreement (Hamidieh et al., 2026) or the base model (Luo et al., 2025) can supply what self-consistency misses.

Medical models and differential diagnosis. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases (Kanjee et al., 2023; McDuff et al., 2025) and in conversation (Tu et al., 2025), and on Phenopacket Store, where they trail the phenotype-driven tool Exomiser (Reese et al., 2026; Chimirri et al., 2025); base models have been left out of clinical evaluations for failing to follow instructions (Hager et al., 2024), the format problem our worked examples address. Medical post-training includes verifier-guided SFT and PPO (Chen et al., 2025), rule-based RL (Liu et al., 2025a) and reasoning distillation (Huang et al., 2026); RL and reward models tend to favor confident answers (Leng et al., 2025; Damani et al., 2026; Mei et al., 2026). Fansi Tchango et al. (2022a) score predicted differentials against DDXPlus references as sets; Puri et al. (2026) show on DDXPlus with Qwen3-8B that single-answer RL collapses sampled answers and train models to output several answers per completion; their single-answer RL model's prompted list already covers slightly more than its samples (0.67 against 0.62), though not for their calibration-rewarded model (0.49 against 0.65). Concurrent work scores differentials over k attempts against physicians' differentials and finds that the best configuration for one draw is often not the best at k (Le Bronnec and Yokota, 2026). Relative to Jang et al. (2026) (who compare stated and sampled distributions on synthetic and survey targets with two of our pairs), Puri et al. (2026) and Karouzos et al. (2026), we add a clinical reference distribution with a true diagnosis, real case reports scored against their authors' own differentials, must-not-miss conditions taken from published sources, a test of whether a post-trained list reproduces its base model's per-case view, prompt, temperature, decoding and chance controls, and the consequences for voting and error signals.

3. Setup

Tasks. **DDXPlus** (Fansi Tchango et al., 2022b): synthetic patients from a rule-based medical knowledge base, each with a true pathology and a reference differential, the engine's posterior over 49 conditions. We use 980 test patients (20 per condition) rendered by template as questionnaire-style vignettes without exam or test results (example in Appendix B). The prompt lists the 49 conditions and asks for the five most likely, with probabilities, as JSON, so this is a closed-set task. **MedCaseReasoning** (Wu et al., 2025b): 897 presentations written from open-access case reports (2005–2025), each with the report's final diagnosis and, in its reasoning notes, the alternatives the authors considered. The differential is open-ended; a judge model from a family we do not evaluate on these cases decides whether a name matches the final diagnosis or one of the authors' alternatives, and a second one gives a strict grading (Appendix B). **Phenopacket Store** (Danis et al., 2025): 1,401 published rare-disease cases (749 diseases) as phenotype vignettes, graded via Mondo (Appendix B).

Models and sampling. Qwen2.5-7B, Qwen3-8B, Qwen2.5-32B and Mistral-7B-v0.3, each with its official post-trained version, and, added later, Qwen2.5-72B and the medical MedGemma-4B; OLMo-3-1025-7B and OLMo-3-1125-32B with the SFT, DPO and final RLVR (Instruct) checkpoints of OLMo 3 7B and OLMo 3.1 32B, plus OLMo-3-7B-RL-Zero-General,

trained with RL from the 7B base; and AlphaMed, HuatuoGPT-o1 and m1, fine-tuned from Qwen2.5-7B-Instruct (Table 6). We draw 32 samples per case with vLLM at T=0.7 (top-p 0.95) and, for the 7–8B models, T=1 (top-p 1), each model with its own chat template (base models without one get a post-trained sibling's) and Qwen3's thinking mode off unless stated. A sample's *top answer* is valid if it is one of the 49 conditions and its *list* if it names at least one; invalid answers count as wrong. With this zero-shot prompt base models are often invalid (up to 47% of samples for OLMo and 76% for Mistral), so every final pair is re-run with three worked examples from the training split (Appendix F.1), on which every general-domain model is valid in over 99% of samples.

Readouts. The *sampled* readout is the distribution of top answers over a case's samples: pass@k (Chen et al., 2021), distinct top answers, the reference mass they cover (coverage@32), and total-variation distance (TV) of their empirical distribution to the reference. The *stated* readout is one sample's list, with its probabilities renormalized over the conditions it names (uniform if missing), averaged per case over valid samples: whether it contains the true pathology among its top k (recall@k), its TV to the reference (also with the true pathology removed from both), the reference mass it names, its *conditional* TV (to the reference restricted to the named conditions and renormalized, a scale-free measure of how it allocates probability), DDXPlus's set F1 against the reference's support (DDF1), and whether it names each severe condition (DDXPlus severity 1–2, the dataset's label) that the reference weights at least 0.1. The *single-call answer* is the top answer of a model's first valid sample.

Statistics. DDXPlus has 20 cases per true pathology, so intervals are 95% percentile intervals from a bootstrap that resamples the 49 pathologies with all their cases (10,000 resamples; 50,000 for the headline tests); p-values use the add-one correction. The 112 headline tests (14 metrics fixed in advance, eight zero-shot settings) are primary and Holm-corrected together; every control, stage, few-shot and per-condition analysis is secondary, corrected within its own family (Appendix J), and read as a check on them. Appendix intervals are condition-clustered too (Phenopacket Store: by disease; case reports: by case); only the pass@k curves in the figures use case-level bands. AUROCs use average ranks for ties. A *setting* is one model pair at one temperature and prompt. Analyses added after the headline tests were fixed, mostly in response to reviews, are secondary: the Qwen2.5-72B and MedGemma-4B pairs, the chance correction, the equal-size comparisons, the readout reversal and silent errors, and the case-report regradings. With five to six pipelines, results describe these checkpoints rather than post-training in general.

4. Results

4.1. Post-training collapses the sampled answers

Table 2 compares each base model with its post-trained versions. Final post-trained models give 1.3–2.4 distinct top answers per 32 samples at T=0.7, against 3.8–4.9 (Qwen2.5-72B: 3.0), also when asked for one diagnosis per call instead of a list (1.2–1.4 against 3.5–3.8; Appendix D.3). At 7–8B the post-trained Qwen model is far better on its first answer (+12.6 and +17.0 points at T=0.7, up to +27.5 at T=1), but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 9.7–14.6 points (Figure 2);

at 32B the zero-shot losses (4.9 and 4.0 points) do not survive correction. Because base models often answer the zero-shot prompt invalidly, we also compare each base model given worked examples with its post-trained model at its better prompt: post-training then costs 13.2–29.9 points of pass@32 in all six final pairs, including both 32B pairs and Mistral (Appendices F.1–F.8). Part of pass@32 is chance: scored against another patient's diagnosis, base models' samples still reach 0.05–0.19 and post-trained ones 0.01–0.04, and after subtracting these the loss at each model's best prompt is 5.9–17.6 points, significant in four of six pairs (not for Qwen3-8B, 7.7, or Mistral, 5.9) (Appendix F.5). Pooled across pairs, the zero-shot loss is 11.9 points [8.5, 15.4] (Table 47). Two pairs added later collapse too: Qwen2.5-72B (−12.3 best-prompt pass@32) and, with worked examples, MedGemma-4B (−15.0, n.s. after chance correction; Appendices D.5, F.9).

Two controls separate sharper sampling and prompt format from lost breadth. A hotter post-trained model is not a base model: along a temperature frontier with worked examples (Appendix F.3), OLMo-3-7B-Instruct and both 32B post-trained models reach less pass@32 than their base models at the same pass@1 (by 6.6–31.9 points; 4.5, n.s., for OLMo 3.1 32B at T=1.3), and at 32B the base model at T=0.4 beats its post-trained model on both pass@1 and pass@32 at every temperature. Only Qwen2.5-7B-Instruct reaches its base model's frontier. And much of the first-answer gain is prompt format: with the same worked examples the sampled pass@1 gain falls to 1.6–5.2 points in four of six pairs, and decoded greedily, post-training significantly raises top-1 only for Qwen3-8B and Mistral, lowers it for Qwen2.5-32B and OLMo 3 7B under two of the three pairings, and is equivalent to the base model within 5 points in no pair (Appendix F.3). Qwen3-8B with thinking on reaches at least its base model's pass@32 (0.815 against 0.732), though its samples stay narrow (1.76 distinct answers; Appendix G.1).

4.2. The readouts reverse: one stated list keeps its coverage

Post-training reverses which readout finds the true pathology (Figure 1). For every base model, 32 samples name it more often than the first valid stated list (by 4.5–28.6 points), and two to eight lists are needed to match them; for every post-trained model, the first list already names it more often than all 32 samples (by 2.1–19.6 points, significant in seven of eight pairs), so post-training moves the difference by 17–41 points in every pair. The reversal is the collapse seen from the list's side, not a better ranking: the list names five conditions and the post-trained samples 1.3–2.4, and at equal size the two post-trained readouts are within 5.2 points, samples ahead at 32B, while base models' samples beat their list by 8.5–14.6 (Appendix E.2); the list wins because it still names alternatives the samples no longer reach (Appendix E.6). Zero-shot, for 7–20% of patients a post-trained model's first list names the true pathology and none of its 32 samples does (about 3% for base models). After subtracting what each readout reaches against another patient's diagnosis (post-trained list 0.07–0.09, its samples 0.01–0.04), base models' samples stay significantly ahead in all eight pairs, while the post-trained list's lead stays significant for Qwen3-8B (+12.4), Qwen2.5-7B (+7.9) and MedGemma-4B (+6.6) only. On the case reports the reversal holds for the authors' alternatives in three of five pairs, but not for the final diagnosis, which 32 post-trained samples contain more often than one list in four of five pairs (Appendix H.3).

Comparison	Model	Sampled top answers pass@1	pass@32	Distinct	Cov@32	Drawn from stated lists pass@32	Distinct	Union of lists
Qwen3-8B, T=0.7	base	0.388	**0.732**	**3.77**	**0.323**	0.787	7.80	0.883
	post-trained	**0.559**	0.607	1.27	0.217	**0.812**	4.98	0.847
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**0.407**	**0.834**	9.47	0.923
	post-trained	**0.556**	0.643	1.53	0.236	0.815	5.41	0.866
Qwen2.5-7B, T=0.7	base	0.352	**0.695**	**4.91**	**0.335**	0.773	10.64	0.857
	instruct	**0.478**	0.598	1.82	0.239	**0.793**	6.83	0.874
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**0.422**	0.796	12.07	0.907
	instruct	**0.464**	0.663	2.65	0.290	**0.822**	7.81	0.921
Qwen2.5-32B, T=0.7	base	0.500	**0.795**	**3.78**	**0.352**	0.856	8.76	0.937
	instruct	**0.604**	0.746	1.68	0.276	**0.883**	6.51	0.932
OLMo 3 7B, T=0.7	base	0.178	**0.705**	**4.93**	**0.351**	0.700	8.54	0.876
	SFT+DPO+RLVR	0.265	0.422	2.37	0.195	0.574	5.59	0.771
	RL-Zero (from base)	**0.295**	0.572	4.06	0.286	**0.704**	9.62	0.830
OLMo 3.1 32B, T=0.7	base	0.326	**0.796**	**4.32**	**0.347**	0.840	8.09	0.942
	SFT+DPO+RLVR	**0.560**	0.756	2.15	0.283	**0.847**	6.71	0.906
Chance	uniform guesser	0.020	0.48	23.6	0.47			
	reference-prior guesser	—	0.44	21.9	0.56			

Table 2: Sampled readout on DDXPlus (980 cases, 32 samples). Distinct: distinct top answers per case; Cov@32: reference mass they cover. *Drawn from stated lists*: each sample's top answer replaced by a draw from its own stated list; *union of lists*: the union of the stated lists of all valid samples contains the true pathology, the ceiling for drawn pass@32. Bold: better model of each pair on the sampled columns and drawn pass@32 (OLMo: of the block; the SFT and DPO stages are in Tables 11 and 12). Invalid answers count as wrong, which mostly penalizes base models. Chance: guessers that ignore the patient, drawing uniformly from the 49 conditions or from the reference differential averaged over test cases.

Within one call, a post-trained list names as many conditions as a base-model list (about five) but more of the right ones: 5–9 points more reference mass and 7–13 points more of the conditions the reference warrants, for every model but OLMo 3 7B (Table 15). Across calls the stated readout narrows too, because post-trained lists barely vary (TV 0.10–0.33 to the mean of the model's other lists, against 0.37–0.69 for base lists): 32 post-trained lists together name 4.6–10.0 fewer conditions than 32 base-model lists and 7–14 points fewer of the warranted ones, though they still cover more than twice the reference mass of the same model's 32 samples. Each post-trained list stays closer to its base model's mean list for the same patient than for other patients with the same diagnosis (all 15 settings), while post-trained lists also converge on one another (Tables 38, 33).

Whether the list improves on the base model's depends on the model (Table 13 in Appendix E). Zero-shot, for every Qwen model and OLMo 3.1 32B, one post-trained list contains the true pathology among its top five 12–16 points more often at T=0.7 (up to 24 at T=1), is closer to the reference (TV −0.043 to −0.062, closing 8–12% of the gap between a patient-blind list and the reference's own top five; up to −0.088 at T=1) and allocates probability among the named conditions more like it (conditional TV −0.025 to −0.078); OLMo 3 7B's lists do not improve. The gains survive complete lists, removing the true pathology (except Qwen3-8B at T=0.7), natural prevalence and free-length lists (Appendices E.4, F.6, E.3). With worked examples they persist for the smaller Qwen models

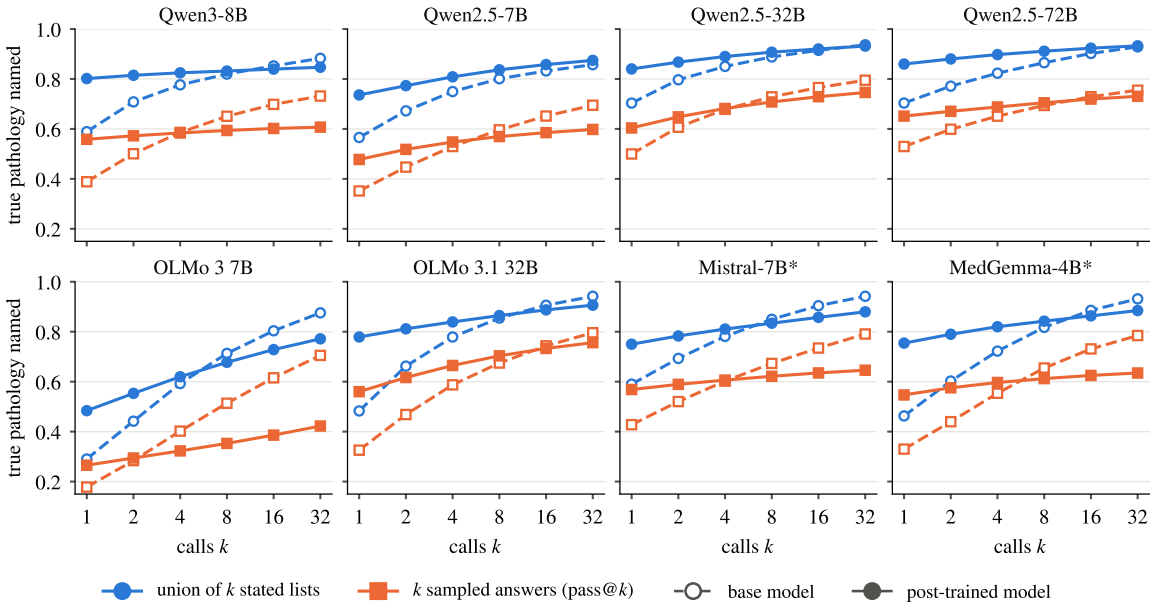


Figure 1: Readout reversal on DDXPlus (T=0.7, 980 cases; zero-shot, *: with worked examples). Share of patients whose true pathology is named by the top answers of k samples (pass@k, orange squares) or anywhere in the union of the stated lists of k calls (blue circles; invalid calls name nothing; mean over 20 orders), for the base (hollow, dashed) and post-trained (filled, solid) model. A base model's 32 samples beat one of its lists; a post-trained model's one list beats its 32 samples.

and Mistral, persist in TV but not recall for Qwen2.5-32B, and weaken for OLMo 3.1 32B; pooled across pairs recall rises in every prompt setting, and the stated list gains more for Qwen than for the other families (zero-shot recall@5 by 6.8 points [3.6, 10.0]; Table 47). The gain is not confined to Qwen: it also holds for OLMo 3.1 32B, Mistral and most strongly MedGemma-4B (recall@5 +27.1 with worked examples; Appendices F.8, F.9). Averaging calls closes most of the gap: the mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700; diamonds in Appendix Figure 4a), and with five lists on both sides post-training is closer in only four of eleven settings (Appendix G.2).

Severe conditions. Post-trained lists name more of the severe conditions the reference warrants in all eight zero-shot settings (+6.0 to +13.8 points in seven, six of them surviving correction; OLMo 3.1 32B +2.2, n.s.), and, in seven of eight settings, more of the must-not-miss conditions taken from diagnostic-harm data and the chest-pain guideline (Newman-Toker et al., 2024; Gulati et al., 2021), but one list still names only 0.32–0.47 of the warranted severe conditions, and some models lose specific ones, most often acute pulmonary edema (Appendices E.4, F.4). The vignettes lack the ECG and biomarkers that separate the coronary syndromes, so these shifts track the reference engine rather than clinical correctness.

4.3. Tools that read the differential from samples lose value

Table 3 summarizes what the collapse does to tools built on samples. On DDXPlus, majority voting over 32 samples raises base models' accuracy by 3.4–8.4 points but post-trained models' by at most 1.9, while models trained to reason before answering and a chain-of-thought prompt still gain 3.8–4.8 (thinking mode +1.5, n.s.; Table 40). Agreement, the share of 31 further samples with the same top answer as the single-call answer, ranks that answer's correctness less well after post-training in every pair (by 0.10 $[0.06, 0.13]$ pooled; individually significant for the Qwen pairs), also where the samples are not unanimous (Appendix F.7), consistent with Liu (2026). Agreement cannot flag an error the model repeats: every other valid sample repeats 19–61% of post-trained models' wrong single-call answers, against 1–9% for base models, and the stated probability is no remedy, since referring the 20% of cases where it is lowest catches only 5–29% of these silent errors (eight pairs), about what random referral would or less (Appendix E.6). Which signal is better then depends on the model (zero-shot, the stated probability for Qwen3-8B, +0.079 $[0.011, 0.142]$; agreement for Qwen2.5-7B-Instruct, OLMo-3.1-32B-Instruct and thinking mode), and for post-trained models a logistic combination fitted on 100 labeled cases adds 0.030 $[0.013, 0.044]$ held-out AUROC over agreement (Appendix G.3).

On 897 presentations written from published case reports, scored against each report's final diagnosis and the alternatives its authors considered (Appendix H.3), the split replicates. After post-training the 32 samples cover 8–14 points fewer of the authors' alternatives for the Qwen and OLMo 3.1 32B pairs (7–13 when every judge acceptance that shares no content word with its target is discarded; OLMo 3 7B n.s.), while one list covers as many as before (equivalent within 5 points) and contains the final diagnosis 2–8 points more often (OLMo 3 7B n.s.). With a closed list of ten candidates graded by name, all four pairs show the split, more sharply (Appendix H.4). Voting helps little for any model; after post-training the stated probability is the better error signal for Qwen3-8B and OLMo 3.1 32B (and for Qwen2.5-32B unless paraphrases are merged) and not significantly different from agreement for the other two. pass@32 for the final diagnosis falls significantly only for Qwen3-8B, and both OLMo Instruct models' answers vary more than their base models' (much less so for OLMo 3.1 32B once paraphrases are merged). The judges agree on 97.5% of names (Cohen's κ 0.77) and were checked against string matches.

5. Discussion and limitations

Jang et al. (2026) show that instruction-tuned models cannot sample from distributions they can describe. In differential diagnosis we find the same split: a single post-trained list keeps its coverage of the differential while varying far less between calls, and the sampled answers collapse. Evaluations that read the differential from samples (pass@k at large k, diversity, self-consistency, agreement-based uncertainty) therefore report a loss that a clinician reading one answer would not see, and error signals that work for base models and reasoning modes work less well for post-trained models answering directly. In distance to the reference, one post-trained call adds little over five averaged base calls, and with a shared prompt and greedy decoding its first-answer gain is not significant in four of six pairs, though equivalent in none. The lost breadth is in the list: one post-trained list beats 32 of the model's samples, and the union of a few lists covers more (Figure 1). Drawing answers from the list instead is

	DDXPlus (980 patients)			Published case reports (897)		
Pair	Vote@32 gain base / post	Agreement AUROC base → post	Stated − agree post	Authors' alternatives covered 32 samples	one list	Stated − agree post
Qwen3-8B	+3.4 / +0.3	0.810 → 0.655	+0.079	0.330 → 0.190	0.300 → 0.296‡	+0.085
Qwen2.5-7B	+5.2 / +0.8	0.811 → 0.702	−0.131	0.376 → 0.288	0.286 → 0.281‡	+0.030†
Qwen2.5-32B	+7.0 / +1.9	0.797 → 0.721	−0.011†	0.331 → 0.247	0.319 → 0.318‡	+0.062
OLMo 3 7B	+4.0 / −0.2	0.802 → 0.731†	+0.064†	0.351 → 0.322†	0.248 → 0.236‡	+0.023†
OLMo 3.1 32B	+8.4 / +0.9	0.831 → 0.772†	−0.077	0.359 → 0.266	0.293 → 0.294‡	+0.114

Table 3: Consequences of the collapse (T=0.7, zero-shot, 32 samples). *Vote@32 gain*: accuracy of the majority answer minus that of a single call, in points. *Agreement AUROC*: how well agreement among 31 further samples ranks the correctness of the single-call answer. *Stated − agree*: AUROC of the single-call answer's stated probability minus that of agreement, for the post-trained model. *Authors' alternatives covered*: share of the alternatives the authors considered that any of the 32 sampled answers, or one stated list, names (judge-graded). †: not significant (Holm-corrected headline test, condition-clustered interval or case bootstrap, by column; Appendices J, H.3; vote gains not tested); ‡: equivalent within 5 points. Full results: Tables 40, 43.

not significantly below the worked-example base model's chance-corrected pass@32 for the Qwen 7–8B and 72B models but is at 32B (about 8 points) and for OLMo 3 7B (22), and is equivalent within 5 points only for Qwen3-8B before chance correction (Appendix E.4). On published case reports the split holds for the authors' alternatives, but the samples still find the final diagnosis more often, so the list replaces sampling as the source of alternatives, not of the best answer. We suggest that studies of diagnostic breadth report which readout they use and whether the model reasons first, that systems needing several candidates read them from one stated list, or the union of a few, rather than from repeated samples, and that the error signal be chosen on a small labeled sample from the target population.

Limitations. DDXPlus is synthetic and closed-set: its reference differential is a rule-based engine's posterior over questionnaire answers without exam or test results, sometimes clinically implausible (Appendix B), our 980-case test sample is class-balanced, and the prompt lists all 49 conditions, so our metrics measure agreement with that engine, not clinical correctness, and no clinician reviewed lists or references. Severity labels come from the dataset or a published list, and the prompt asks for the most likely conditions, not must-not-miss ones. The few-shot controls use one set of example patients, the matched-temperature control covers the two 7–8B Qwen models only, and the free-length check the Qwen pairs only (OLMo base models rarely return a usable free-length list). We study open 4–72B models from four families, one reasoning mode and no frontier models. Case-report grading and extraction rely on language models checked only against string matches, and published cases may appear in pretraining data; Phenopacket accuracy is near the floor.

Statement on AI Use. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to draft and edit text, and to check citations. The authors designed the study, reviewed all code, results and text, and take responsibility for the content.

References

Shivam Agarwal, Zimin Zhang, Lifan Yuan, Jiawei Han, and Hao Peng. The unreasonable effectiveness of entropy minimization in LLM reasoning. In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

Kasra Arabi, Virginia Smith, and Chhavi Yadav. Sample what you say: Aligning language models to sample the distributions they state, 2026.

Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. Towards medical complex reasoning with LLMs through medical verifiable problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.751. arXiv version titled "HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs".

Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code, 2021.

Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, et al. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

Leonardo Chimirri, J. Harry Caufield, Yasemin Bridges, Nicolas Matentzoglu, Michael Gargano, et al. Consistent performance of large language models in rare disease diagnosis across ten languages and 4917 cases. *eBioMedicine*, 121:105957, 2025. doi: 10.1016/j.ebiom.2025.105957.

Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, et al. The entropy mechanism of reinforcement learning for reasoning language models. *arXiv preprint arXiv:2505.22617*, 2025.

Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations (ICLR)*, 2026.

Daniel Danis, Michael J. Bamshad, Yasemin Bridges, Andrés Caballero-Oteyza, Pilar Cacheiro, Leigh C. Carmody, Leonardo Chimirri, Jessica X. Chong, Ben Coleman, Raymond Dalgleish, Peter J. Freeman, Adam S. L. Graefe, Tudor Groza, Peter Hansen, Julius O. B. Jacobsen, Adam Klocperk, Maaike Kusters, Markus S. Ladewig, Allison J. Marcello, Teresa Mattina, Christopher J. Mungall, Monica C. Munoz-Torres, Justin T. Reese, Filip Rehburg, Bárbara C. S. Reis, Catharina Schuetz, Damian Smedley, Timmy Strauss, Jagadish Chandrabose Sundaramurthi, Sylvia Thun, Kyran Wissink, John F. Wagstaff, David Zocche, Melissa A. Haendel, and Peter N. Robinson. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. *Human Genetics and Genomics Advances*, 6(1):100371, 2025. doi: 10.1016/j.xhgg.2024.100371.

Marius Dragoi, Ioana Pintilie, Florin Gogianu, and Florin Brad. Beyond pass@k: Breadth-depth metrics for reasoning boundaries, 2025.

Arsene Fansi Tchango, Rishab Goel, Julien Martel, Zhi Wen, Gaetan Marceau Caron, and Joumana Ghosn. Towards trustworthy automatic diagnosis systems by emulating doctors' reasoning with deep reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, 2022a.

Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. DDXPlus: A new dataset for automatic medical diagnosis. In *Advances in Neural Information Processing Systems (NeurIPS 2022), Datasets and Benchmarks Track*, 2022b.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

Xiangming Gu, Soham De, Michalis Titsias, Larisa Markeeva, Petar Veličković, and Razvan Pascanu. The illusion of stochasticity in LLMs. In *Conference on Language Modeling (COLM)*, 2026. arXiv:2604.06543.

Martha Gulati, Phillip D. Levy, Debabrata Mukherjee, Ezra Amsterdam, Deepak L. Bhatt, et al. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain. *Circulation*, 144(22):e368–e454, 2021. doi: 10.1161/CIR.0000000000001029.

Paul Hager, Friederike Jungmann, Robbie Holland, Kunal Bhagat, Inga Hubrecht, et al. Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nature Medicine*, 30(9):2613–2622, 2024.

Kimia Hamidieh, Veronika Thost, Walter Gerych, Mikhail Yurochkin, and Marzyeh Ghassemi. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *International Conference on Learning Representations (ICLR)*, 2026.

Audrey Huang, Adam Block, Dylan J. Foster, Dhruv Rohatgi, Cyril Zhang, Max Simchowitz, Jordan T. Ash, and Akshay Krishnamurthy. Self-improvement in language models: The sharpening mechanism. In *International Conference on Learning Representations (ICLR)*, 2025.

Xiaoke Huang, Juncheng Wu, Hui Liu, Xianfeng Tang, and Yuyin Zhou. m1: Unleash the potential of test-time scaling for medical reasoning with large language models. In *Proceedings of the Fifth Machine Learning for Health Symposium (ML4H 2025)*, volume 297 of *Proceedings of Machine Learning Research*, pages 369–383. PMLR, 2026.

Chaemin Jang, Jihee Kim, and Dongman Lee. Instruction-tuned language models cannot sample from distributions they can describe. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 48757–48777, 2026.

Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Zahir Kanjee, Byron Crowe, and Adam Rodman. Accuracy of a generative artificial intelligence model in a complex diagnostic challenge. *JAMA*, 330(1):78–80, 2023. doi: 10.1001/jama.2023.8288.

Aayush Karan and Yilun Du. Reasoning with sampling: Your base model is smarter than you think. In *International Conference on Learning Representations (ICLR)*, 2026.

Constantinos Karouzos, Xingwei Tan, and Nikolaos Aletras. Where does output diversity collapse in post-training?, 2026.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.

Thom Lake, Eunsol Choi, and Greg Durrett. From distributional to overton pluralism: Investigating large language model alignment. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 6794–6814, 2025.

Florian Le Bronnec and Rio Yokota. Evaluating multiple LLM generations with validated task coverage, 2026.

Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

Bill Yuchen Lin, Abhilasha Ravichander, Ximing Lu, Nouha Dziri, Melanie Sclar, Khyathi Chandu, Chandra Bhagavatula, and Yejin Choi. The unlocking spell on base LLMs: Rethinking alignment via in-context learning. In *International Conference on Learning Representations (ICLR)*, 2024.

Che Liu, Haozhe Wang, Jiazhen Pan, Zhongwei Wan, Yong Dai, Fangzhen Lin, Wenjia Bai, Daniel Rueckert, and Rossella Arcucci. Beyond distillation: Pushing the limits of medical LLM reasoning with minimalist rule-based RL, 2025a. NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. ProRL: Prolonged reinforcement learning expands reasoning boundaries in large language models. In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025b.

Mingyi Liu. The alignment tax: Response homogenization in aligned LLMs and its implications for uncertainty estimation, 2026.

Beier Luo, Shuoyuan Wang, Sharon Li, and Hongxin Wei. Your pre-trained LLM is secretly an unsupervised confidence calibrator. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2505.16690.

Qing Lyu, Kumar Shridhar, Chaitanya Malaviya, Li Zhang, Yanai Elazar, Niket Tandon, Marianna Apidianaki, Mrinmaya Sachan, and Chris Callison-Burch. Calibrating large language models with sample consistency. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 19260–19268, 2025. doi: 10.1609/aaai.v39i18.34120.

Daniel McDuff, Mike Schaekermann, Tao Tu, Anil Palepu, Amy Wang, Jake Garrison, Karan Singhal, Yash Sharma, Shekoofeh Azizi, Kavita Kulkarni, et al. Towards accurate differential diagnosis with large language models. *Nature*, 642(8067):451–457, 2025. doi: 10.1038/s41586-025-08869-4.

Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don't know? In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 3408–3458, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-eacl.178.

Nicole Meister, Carlos Guestrin, and Tatsunori Hashimoto. Benchmarking distributional alignment of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 24–49, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.naacl-long.2.

Sonia Krishna Murthy, Tomer Ullman, and Jennifer Hu. One fish, two fish, but not the whole sea: Alignment reduces language models' conceptual diversity. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for*

Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 11241–11258, 2025. doi: 10.18653/v1/2025.naacl-long.561.

David E. Newman-Toker, Najlla Nassery, Adam C. Schaffer, Chihwen Winnie Yu-Moe, Gwendolyn D. Clemens, et al. Burden of serious harms from diagnostic error in the USA. *BMJ Quality & Safety*, 33(2):109–120, 2024. doi: 10.1136/bmjqs-2021-014130.

Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, Hangoo Kang, Azalia Mirhoseini, and Sharon Li. Sharpening tax in post-training, 2026.

OpenAI. GPT-4 technical report, 2023.

Max Peeperkorn, Tom Kouwenhoven, Dan Brown, and Anna Jordanous. Mind the gap: Conformative decoding to improve output diversity of instruction-tuned large language models, 2025.

Archiki Prasad, Weizhe Yuan, Richard Yuanzhe Pang, Jing Xu, Maryam Fazel-Zarandi, Mohit Bansal, Sainbayar Sukhbaatar, Jason Weston, and Jane Yu. Self-consistency preference optimization. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2411.04109.

Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, and Yoon Kim. Escaping the mode: Multi-answer reinforcement learning in LMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 100127–100146. PMLR, 2026. arXiv version titled "Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models".

Justin T. Reese, Leonardo Chimirri, Yasemin Bridges, Daniel Danis, J. Harry Caufield, Michael A. Gargano, Carlo Kroll, Andrew Schmeder, Fengchen Liu, Kyran Wissink, Julie A. McMurry, et al. Systematic benchmarking demonstrates large language models have not reached the diagnostic accuracy of traditional rare-disease decision support tools. *European Journal of Human Genetics*, 34(4):498–504, 2026. doi: 10.1038/s41431-026-02054-5.

Zhiyao Ren, Yibing Zhan, Siyuan Liang, Guozheng Ma, Baosheng Yu, and Dacheng Tao. Towards reliable medical LLMs: Benchmarking and enhancing confidence estimation of large language models in medical consultation, 2026.

Thomas Savage, John Wang, Robert Gallo, Abdessalem Boukil, Vishwesh Patel, Seyed Amir Ahmad Safavi-Naini, Ali Soroush, and Jonathan H. Chen. Large language model uncertainty proxies: discrimination and calibration for medical diagnosis and treatment. *Journal of the American Medical Informatics Association*, 32(1):139–149, 2025. doi: 10.1093/jamia/ocae254.

Yuda Song, Julia Kempe, and Remi Munos. Outcome-based exploration for LLM reasoning, 2025.

Jacob Mitchell Springer, Madhu Advani, Lukas Aichberger, Arwen Bradley, Eran Malach, Omid Saremi, Sinead Williamson, Preetum Nakkiran, Etai Littwin, and Aditi Raghunathan. Annotations mitigate post-training mode collapse. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.

Hexiang Tan, Fei Sun, Sha Liu, Du Su, Qi Cao, Xin Chen, Jingang Wang, Xunliang Cai, Yuanzhuo Wang, Huawei Shen, and Xueqi Cheng. Too consistent to detect: A study of self-consistent errors in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2025. arXiv:2505.17656.

Amir Taubenfeld, Tom Sheffer, Eran Ofek, Amir Feder, Ariel Goldstein, Zorik Gekhman, and Gal Yona. Confidence improves self-consistency in LLMs. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. arXiv:2502.06233.

Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, et al. Olmo 3, 2025.

Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Sergey Troshin, Irina Saparina, Antske Fokkens, and Vlad Niculae. Asking a language model for diverse responses. In *Proceedings of the 2nd Workshop on Uncertainty-Aware NLP (UncertaiNLP 2025)*, pages 66–72. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.uncertainlp-main.8.

Tao Tu, Mike Schaekermann, Anil Palepu, Khaled Saab, Jan Freyberg, et al. Towards conversational diagnostic artificial intelligence. *Nature*, 642(8067):442–450, 2025.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. In *Conference on Language Modeling (COLM)*, 2025.

Fang Wu, Weihao Xuan, Danny Xing, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin, 2025a.

Kevin Wu, Eric Wu, Rahul Thapa, Kevin Wei, Angela Zhang, Arvind Suresh, Jacqueline J. Tao, Min Woo Sun, Alejandro Lozano, and James Zou. MedCaseReasoning: Evaluating and learning diagnostic reasoning from clinical case reports. *arXiv preprint arXiv:2505.11733*, 2025b.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations (ICLR)*, 2024.

Chenghao Yang, Sida Li, and Ari Holtzman. LLM probability concentration: How alignment shrinks the generative horizon. *Transactions on Machine Learning Research*, 2026. arXiv:2506.17871.

Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152. PMLR, 2026.

Yiming Zhang, Avi Schwarzschild, Nicholas Carlini, Zico Kolter, and Daphne Ippolito. Forcing diffuse distributions out of language models. In *Conference on Language Modeling (COLM)*, 2024.

Minda Zhao, Yilun Du, and Mengyu Wang. Large language models are bad dice players: LLMs struggle to generate random numbers from statistical distributions. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. arXiv:2601.05414.

Chiwei Zhu, Benfeng Xu, Quan Wang, Yongdong Zhang, and Zhendong Mao. On the calibration of large language models and alignment. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2311.13240.

Appendix A. Claims and the evidence for them

Table 4 lists each claim with its primary evidence, the checks under which it holds and those under which it weakens or fails. For claims (b), (e) and (g) the primary evidence comes from the 112 headline tests (§3); for the others it is the analysis named in the table, and everything else is a secondary check, and Table 21 gives the same information control by control.

Claim	Primary evidence	Holds under	Weaker or fails under
(a) A post-trained list stays close to the base model's view of each patient and varies little between calls (§4.2)	A post-trained list is closer to its own base model's mean list for the same patient than to the means for other patients with the same diagnosis, in all 15 settings (TV −0.028 to −0.076); post-trained lists vary 0.10–0.33 between calls, base lists 0.37–0.69 (Table 38)	worked examples; thinking mode; against other families' base models (−0.03 to −0.14); for three of five pairs, against every other base model (Table 33)	about as close to the nearest other post-trained model; Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct not closest to their own base
(b) Sampled answers collapse (§4.1)	pass@32 −9.7 to −14.6 points for the 7–8B Qwen pairs and −28.3 for OLMo 3 7B zero-shot; 1.3–2.4 distinct answers against 3.8–4.9 (Table 2)	worked examples of either kind (13–30 points in all six pairs); temperature frontier at matched pass@1 (OLMo 3 7B, both 32B pairs); free-length lists; one diagnosis per call; closed-set case reports; Mistral; MedGemma-4B and 32B, 72B with worked examples	32B and 72B zero-shot (n.s.); chance-corrected at the best prompt, Qwen3-8B and Mistral (n.s.); MedGemma-4B after chance correction (n.s.); Qwen2.5-7B-Instruct at T=1.3; thinking mode; RL-Zero at T=1; seeded pick from the model's own list (4 of 5 models)
(c) Much of post-training's first-answer gain is prompt format (§4.1)	With shared worked examples the sampled pass@1 gain is 1.6–5.2 points in four of six pairs; decoded greedily, post-training raises top-1 significantly in at most two of six pairs under either matched prompt (Table 28)	both matched pairings; both 32B pairs	Qwen3-8B (+10.7 and +13.7); Mistral with shared examples (+10.8); no pair equivalent within 5 points; top-1 lower for Qwen2.5-32B and OLMo 3 7B in two of three pairings
(d) One stated list improves robustly for the 7–8B and 72B Qwen models (§4.2)	Recall@5 +12 to +24 points, TV −0.043 to −0.088, conditional TV −0.025 to −0.078 (Table 13, Table 46)	complete lists; correct and wrong top answers; true pathology removed (not Qwen3-8B, T=0.7); free-length lists; worked examples for Qwen3-8B and Qwen2.5-7B; Qwen2.5-72B; MedGemma-4B (not cond. TV); thinking mode	OLMo 3 7B (no gain); OLMo 3.1 32B and Qwen2.5-32B recall with worked examples; five averaged base lists match one post-trained list
(e) Severe conditions are named more often on average, but specific ones are lost (§4.2)	Named more often in all eight zero-shot settings (+6.0 to +13.8 points in seven, six surviving correction; OLMo 3.1 32B +2.2, n.s.); 2–7 of 17 conditions per pair named significantly less often (Table 17)	published must-not-miss sets (+9 to +24 points in seven of eight settings)	worked examples (mean falls for Qwen2.5-32B and OLMo 3 7B); averaged lists; with published sets only OLMo, Mistral and few-shot Qwen2.5 lose specific conditions

Table 4: The claims and their evidence. Ranges are post-trained minus base; "n.s.": interval includes zero or test does not survive correction.

Claim	Primary evidence	Holds under	Weaker or fails under
(f) Asking for must-not-miss conditions raises coverage, partly by lengthening the list (§F.3, F.4)	+6 to +15 points of warranted severe conditions named (Table 30)	published must-not-miss sets (+15 to +26 points; 8–20 beyond two lists)	costs one to two more unwarranted severe conditions per list
(g) Voting stops helping and agreement weakens as an error signal (§4.3)	Voting +3.4 to +8.4 points for base models, at most +1.9 for post-trained ones; agreement's AUROC falls in every family (Table 40)	worked examples; Mistral; Qwen2.5-72B; MedGemma-4B; one diagnosis per call; rare-disease cases	case reports (voting gains small for every model); models that reason first; thinking mode; Qwen2.5-7B-Instruct at T=1.3; agreement drop n.s. for OLMo
(h) The better error signal depends on model and prompt; combining helps post-trained models modestly (§4.3)	Held-out audit over 56 model–setting combinations: logistic combination +0.030 $[0.013, 0.044]$ over 31-sample agreement for post-trained models, rank average +0.019 (n.s.) (§G.3)	worked examples (stated p beats 5-sample agreement for all five post-trained models, 31-sample agreement for two)	base models and thinking mode (agreement alone as good or better); rare-disease cases (agreement better); on open-ended case reports the stated probability is at least as good after post-training for two of four models and not significantly worse for the others (closed-set: agreement better for Qwen2.5-7B-Instruct)
(i) On published case reports the sampled differential narrows against the authors' own alternatives while one stated list does not (§4.3)	Samples cover 8–14 points fewer of the authors' alternatives for the Qwen and OLMo 3.1 32B pairs; one list equivalent within 5 points; recall of the final diagnosis +2 to +8 (Table 43)	strict grading (judges agree on 97.5%, κ 0.77); closed candidate list graded by name (all four pairs)	OLMo 3 7B n.s.; pass@32 loss significant only for Qwen3-8B; both OLMo Instruct models' answers vary more (+0.7 distinct answers for OLMo 3.1 32B once matched paraphrases are merged)
(j) The narrowing can be undone from the stated list (§E.5)	Answers drawn from each call's own list raise pass@32 for every model but the OLMo 3 7B base (Table 2) and are not significantly below the worked-example base for the Qwen 7–8B and 72B models (chance-corrected; 8 points short at 32B and 22 for OLMo 3 7B; equivalent within 5 points only for Qwen3-8B before chance correction); asked to draw from its own list with a supplied random number, four of five post-trained models are no longer significantly below their base model's pass@32 (Table 19)	drawing done outside the model; in-context for both Qwen2.5 and both OLMo models	Qwen3-8B's picks stay narrow; pass@1 −20 points for the 32B models; agreement not restored
(k) Post-training reverses which readout finds the true diagnosis (§4.2)	Base models' 32 samples beat their first list by 4.5–28.6 points, post-trained models' first list beats their 32 samples by 2.1–19.6; the difference moves by 17–41 points in all eight pairs (Table 20, Figure 1)	union of k lists against pass@k at every k; authors' alternatives on case reports (three of five pairs)	chance-corrected, the post-trained lead is significant for three of eight models; final diagnosis on case reports (samples ahead in four of five)
(l) Agreement cannot flag post-trained models' repeated errors (§4.3)	19–61% of post-trained models' wrong single-call answers are repeated by every other sample, against 1–9% for base models (Table 20)	all eight pairs (6.6–25.1 points more of all cases)	the stated probability flags only 5–29% of them at 20% referral, about chance

Table 5: The claims and their evidence (continued from Table 4).

Appendix B. Prompts, vignettes and grading

DDXPlus. Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12; cluster headache is implausible for this patient, a reminder that the reference is an engine's posterior rather than a clinician's differential):

> `A 13-year-old female presents for evaluation. Symptoms (patient answers): Do you have pain somewhere, related to your reason for consulting? Yes. Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead, cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have you lost your sense of smell? Yes. Do you have nasal congestion or a clear runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors: Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your nose? Yes. […]`

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...].` The full-distribution prompt (§E.1) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §D.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

Phenopacket Store. The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

Published case reports. MedCaseReasoning (Wu et al., 2025b) turns open-access PMC case reports (2005–2025) into a presentation, written up to the point where the authors give a differential, and the report's final diagnosis; we use its 897 test cases. The vignette is the presentation as released. System prompt: `You are an experienced physician and diagnostician.` Instructions: `Based on the case above, give your differential diagnosis. List the 5 most likely specific diagnoses, most likely first, with your probability that each is the correct diagnosis. End your answer with a JSON array in exactly this format: [{"diagnosis":`

`"<name>", "probability": <number between 0 and 1>}, ...]`. Every distinct (case, name) pair among all samples of the eight models, after lower-casing and stripping punctuation, is graded once by Mistral-Small-24B-Instruct-2501 at temperature 0 (the Mistral family is evaluated only on DDXPlus, so no judge grades its own family here): `Does the candidate name the same disease as the final diagnosis? Count it as the same if it is a synonym, an abbreviation, or a more specific form of the final diagnosis. Count it as different if it is a broader category, a related but distinct disease, or only a complication, cause or finding. Answer with one word: yes or no.` Phi-4, from a family we do not evaluate, grades the same pairs. The first judge's grading is primary, fixed before the results; the *strict* grading, which counts a name as correct only if both judges do, is a sensitivity analysis. Neither judge was validated against clinicians, only against silver labels from string matching (Appendix H.3); in a smoke test the first was slightly lenient (it accepted "Meningioma" for "Sclerosing meningioma").

Grading open-ended names. Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio ≥ 92). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

Appendix C. Models

Appendix D. The sampled differential in detail

D.1. General post-training trades sampled coverage for top-1 accuracy

Table 7 and Figure 2 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals [+20.3, +34.7] and [+16.0, +29.2]). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, [+8.3, +21.5]) and 10.3 points (Qwen2.5, [+4.2, +17.3]). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.4 points at k=32 ([+6.1, +19.8]). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 [+6.7, +18.8] and a base lead of 9.7 at k=32 [+4.1, +16.2].

D.2. The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.2 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B (Liu et al., 2025a)	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B (Chen et al., 2025)	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K (Huang et al., 2026)	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
Qwen2.5-72B	n/a	none (pretrained)	base
Qwen2.5-72B-Instruct	Qwen2.5-72B	general SFT + RLHF	instruct
Mistral-7B-v0.3	n/a	none (pretrained)	base
Mistral-7B-Instruct-v0.3	Mistral-7B-v0.3	general instruction tuning	instruct
MedGemma-4B-pt	Gemma 3 4B	medical training	medical base
MedGemma-4B-it	MedGemma-4B-pt	medical instruction tuning	medical instruct
OLMo-3-1025-7B (Team Olmo et al., 2025)	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 6: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.715**	**0.407**	0.196
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.719**	**0.422**	0.290
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.136**
Qwen3-8B, T=0.7	base	0.388	**0.732**	**3.77**	**0.78**	**0.750**	**0.323**	0.118
	post-trained	**0.559**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 7: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct, entropy and TV refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability, over samples whose top answer is one of the 49 conditions. Bold: significantly better model (condition-clustered bootstrap, Holm across the table's 18 tests); for ECE, which is not tested, the lower value.

1.87 for the reference differential (Table 7). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decod-

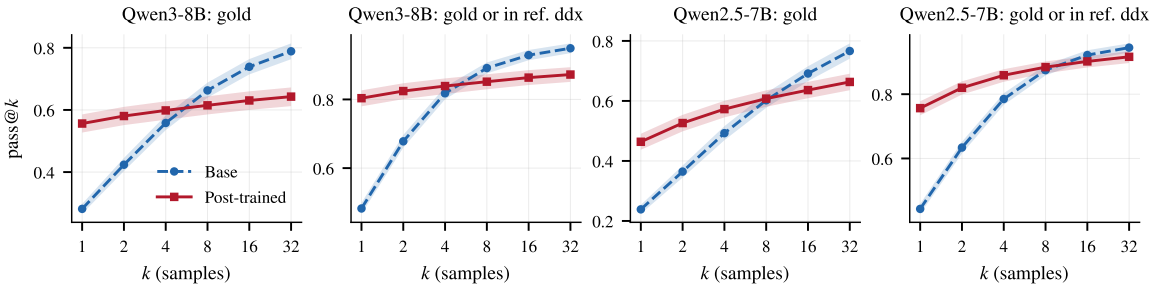


Figure 2: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

ing settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.20 and 0.14 vs. 0.29). This concerns verbalized confidence; alignment is known to worsen the calibration of token probabilities (OpenAI, 2023; Zhu et al., 2023), while verbalized confidence of RLHF models is better calibrated (Tian et al., 2023). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.2).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 8; per condition, Figure 3), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgment; we assess severe conditions on the stated list instead (§E.4). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-22.2, -14.6]$ and -17.0 $[-23.0, -11.5]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95%

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−22.2, −14.6]	−18.1 [−23.7, −12.7]	−24.2 [−29.8, −18.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−23.0, −11.5]	−12.1 [−17.4, −7.9]	−15.0 [−18.2, −11.4]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−13.7, −6.2]	−15.8 [−20.9, −10.8]	−13.4 [−18.6, −8.7]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.31	0.28 → 0.17
	−12.4 [−18.0, −7.4]	−9.1 [−14.2, −4.7]	−10.3 [−14.3, −6.5]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.31 → 0.34	0.17 → 0.21
	+4.3 [−1.2, +10.9]	+2.2 [−0.3, +5.0]	+3.9 [+1.5, +6.7]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.31 → 0.37	0.17 → 0.27
	+8.4 [+3.1, +14.4]	+6.0 [+2.6, +9.7]	+9.5 [+5.7, +14.2]
Instruct → m1, T=0.7	0.28 → 0.41	0.31 → 0.41	0.17 → 0.29
	+13.5 [+6.8, +20.7]	+9.7 [+4.6, +15.4]	+11.4 [+6.9, +16.7]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−5.7, +5.3]	−0.4 [−4.3, +3.6]	−0.4 [−3.8, +3.2]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−5.4, −0.3]	−2.1 [−4.5, +0.2]	−4.3 [−6.7, −1.9]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−4.2, +1.1]	−1.3 [−3.7, +1.0]	−3.0 [−5.9, +0.1]

Table 8: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% condition-clustered intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the severity-1–2 tier; medical post-training does not lower it (HuatuoGPT-o1 and m1 raise it in every tier); our GRPO from the base model loses at most 4.3 points.

for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives. Ruling out acute coronary syndrome first is standard practice for chest pain, but these vignettes lack the ECG and biomarkers that would separate the coronary conditions, so we read this as a concentration of the samples, not as a clinical error.

D.3. Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 10). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt

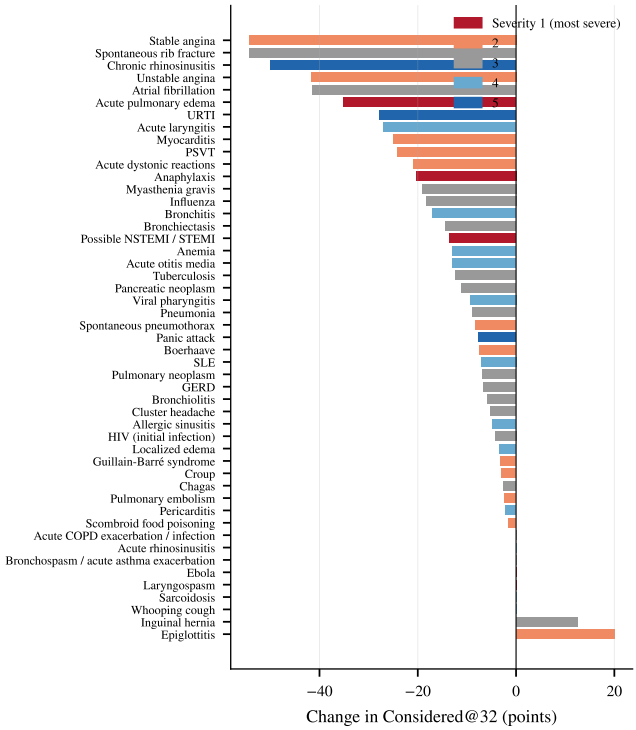


Figure 3: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

reproduces the pattern: the post-trained model gains +23.0 $[+16.2, +30.3]$ points at k=1 and loses 7.1 $[2.4, 12.2]$ points at k=32 (T=0.7; at T=1, +34.2 and −8.7 $[-13.7, -4.0]$), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 (+4.7 $[+0.8, +9.1]$ at k=32) and ties it at T=1 (0.0 $[-3.3, +3.4]$). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§H.1).

One diagnosis per call. The sampled readout takes the first item of each five-item list, so the collapse could come from reading the top of a list. We therefore repeated the two 7–8B Qwen pairs at T=0.7 with a prompt that asks for the single most likely diagnosis, as a one-item JSON list, the format self-consistency and semantic entropy assume (Table 9). The collapse persists at about the same size: post-trained Qwen3-8B and Qwen2.5-7B-Instruct give 1.18 and 1.43 distinct answers per 32 samples, against 3.49 and 3.84 for their

Pair	Model	Invalid	pass@1	pass@32	Distinct	Cov@32	TV samples↓	Vote@32 gain	Agreement AUROC
Qwen3-8B	base	13.9%	0.424	**0.764**	**3.49**	**0.321**	**0.747**	+2.7	**0.824**
	post-trained	0.0%	**0.641**	0.681	1.18	0.222	0.785	$-0.4^{\dagger}$	0.607
Qwen2.5-7B	base	9.4%	0.370	**0.682**	**3.84**	**0.306**	**0.767**	+5.6	**0.822**
	instruct	0.2%	**0.524**	0.598	1.43	0.224	0.790	+1.4	0.683

Table 9: One diagnosis per call (DDXPlus, zero-shot, T=0.7, 980 cases, 32 samples). Invalid: samples whose answer is not one of the 49 conditions (counted as wrong). Vote@32 gain: majority answer minus single call, in points. Agreement AUROC: agreement of 31 further samples with the first answer, for that answer being correct. Bold: better model of each pair; every difference is significant (condition-clustered 95% interval) except where marked † (for Vote@32 gain, the change from base to post-trained; that column is not bolded).

base models (five-item prompt: 1.27 and 1.82 against 3.77 and 4.91), and lose 8.4 points of pass@32 each ($[-13.8, -3.4]$ and $[-13.9, -3.5]$; five-item prompt: 12.4 and 9.7) while gaining 21.7 and 15.3 at pass@1. Agreement among 31 further samples ranks the first answer's correctness less well after post-training (AUROC 0.824 → 0.607 and 0.822 → 0.683), and voting gains the base models 2.7 and 5.6 points and the post-trained models −0.4 and 1.4. Base models give invalid answers in 13.9% and 9.4% of samples. With one item per call the renormalized stated probability is always 1, so it carries no signal here.

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models; samples that name no listed condition at all, the invalid share reported in the tables, are 11.3% and 22.3% (Qwen2.5-7B: 0.9% and 18.4%). Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 $[+5.2, +19.1]$ (Qwen3, T=0.7), +18.7 $[+12.1, +25.4]$ (Qwen3, T=1), +12.2 $[+6.3, +18.3]$ and +16.8 $[+10.9, +22.8]$ points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 $[-25.5, -6.0]$ at k=32 for Qwen3 (105 cases) and −10.2 $[-17.1, -4.2]$ for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

D.4. Which post-training stage narrows the sampled differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 (Team Olmo et al., 2025) releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 11 gives the results.

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.559**	**0.732**	0.607	**3.77**	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	**6.60**	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	**3.72**	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	**5.57**	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	**3.89**	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	**6.31**	2.24

Table 10: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (condition-clustered bootstrap, Holm across the table's 18 tests). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

The coverage loss replicates in OLMo. At T=0.7, every post-trained stage is worse than the base at k=32: by -14.6 $[-20.9, -8.9]$ points after SFT, -9.2 $[-15.6, -3.1]$ after DPO, -28.3 $[-35.6, -21.2]$ after RLVR, and -13.3 $[-19.8, -7.3]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k ($+5.4$ $[+0.3, +10.7]$ points at k=32, and $+10.3$ $[+6.2, +14.5]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-24.8, -13.8]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (-5.9 $[-9.3, -2.8]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-21.1, -10.9]$ points relative to the base, against -7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: -12.7 $[-17.3, -8.2]$ at k=32).

RL applied directly to the base narrows the sampled differential less. At T=1, RL-Zero shows no detectable loss of the base model's pass@32 (-1.8 $[-5.8, +2.1]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[0.9, 7.1]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32 ($[7.3, 19.8]$), about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 2.0% for the later stages. No case has 32 valid base samples, so the all-valid check of §D.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (-1.9 to $+4.0$ points at T=0.7, $+1.3$ to $+8.1$ at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across these pipelines, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§D.5), so which stage

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.265	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 11: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 8. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero at T=1, AlphaMed's base-initialized checkpoint, and our GRPO runs in §I) narrows the sampled differential less, but not always: RL-Zero at T=0.7 loses about as much as SFT. On Phenopacket Store the instruct pipeline also loses sampled coverage: every stage trails the base model at k=32 (-2.0 $[-3.0, -1.0]$ points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 $[-1.4, +0.2]$), from a base model at 4.5% (§H.2).

D.5. Replication at 32B and 72B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 12). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 $[-0.6, 10.4]$ points at pass@32 for Qwen2.5-32B and 4.0 $[1.3, 6.8]$ for OLMo 3.1 32B (neither survives the headline correction), halves the number of distinct answers (3.78 $\rightarrow$ 1.68 and 4.32 $\rightarrow$ 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 $[-9.6, -0.2]$); the OLMo 32B base gives invalid answers on 27% of samples. With few-shot prompts, which nearly remove invalid answers, both losses are three to five times larger (§F.1). The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 -12.1 $[-18.0, -7.0]$, distinct answers -1.39, severe tier -11.9); DPO recovers most of it ($+10.9$ against SFT), and the final RL step is small (-2.8 $[-5.4, -0.5]$ at k=32, -0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. Under this prompt the stated readout improves in both 32B families (§4.2; with few-shot prompts only Qwen2.5-32B's does robustly, §F.1), and voting

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.743**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9
Qwen2.5-72B (base)	0.530	**0.755**	**3.01**	**0.303**	**0.35**	0.740	**0.754**	+4.2
Qwen2.5-72B-Instruct	**0.651**	0.731	1.38	0.252	0.26	**0.681**	0.769	+0.0
base, worked examples	0.674	**0.880**	**3.08**	**0.330**	**0.38**	0.711	**0.721**	+6.7
Instruct, worked examples	**0.711**	0.756	1.19	0.244	0.23	**0.665**	0.762	+0.1

Table 12: 32B and 72B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.2). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.3). Bold: best value per family (Qwen2.5-72B: per prompt). The 72B pair was added after the headline tests were fixed.

stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

Qwen2.5-72B. We added Qwen2.5-72B and Qwen2.5-72B-Instruct after the headline tests were fixed, on DDXPlus at T=0.7 only, zero-shot and with the worked examples of §F.1 (bottom of Table 12); each prompt's 14 headline metrics are Holm-corrected together, outside the 112 primary tests. Its base model answers the zero-shot prompt validly (0.2% invalid samples), so the zero-shot comparison needs no format control. Zero-shot, post-training raises pass@1 by 12.2 points and leaves pass@32 unchanged (-2.4 $[-6.9, 1.8]$; $+0.9$ after chance correction), but it still halves the distinct answers (3.01 → 1.38) and lowers Cov@32 by 0.051 and the severe-tier coverage of the samples by 9.0 points. One list improves as for the smaller Qwen models: recall@5 rises by 15.4 points (0.706 → 0.859), TV falls by 0.059 (the list closes 21% of the gap between the patient-blind list and the reference's top five, against 10% for the base list), and the list names 8.5 points more reference mass and 9.6 points more of the warranted severe conditions. With worked examples the base model reaches pass@32 0.880 and post-training costs 12.3 points (8.8 $[3.6, 14.6]$ after chance correction; Table 32), its pass@1 gain is no longer significant (+3.7), and one list still gains (recall@5 +6.4, TV -0.046). Voting gains the base model 4.2 points zero-shot and 6.7 with worked examples, and the Instruct model 0.0 and 0.1; agreement's AUROC falls from 0.824 to 0.719 zero-shot and from 0.872 to 0.648 with worked examples, and the Instruct model's stated probability is ahead of agreement in both (0.741 and 0.769; not tested). Drawing each zero-shot sample's answer from its own list gives pass@32 0.885, above the base model's 0.755 (breadth in Table 15). As from 7–8B to 32B, the zero-shot pass@32 loss shrinks with scale, but the collapse of the samples, the stated list's gain and the loss of value of voting and agreement remain at 72B.

Comparison (T=0.7)	Model	TV to reference↓ One list	Samples	List w/o true	List TV skill	Reference mass named	Cond. TV↓	DDF1	Recall @5	Severe named
Qwen3-8B	base	0.743	**0.750**	0.833	9.6%	0.334	0.492	0.276	0.666	0.347
	post-trained	**0.700**	0.793	**0.815**	17.8%	**0.396**	**0.431**	**0.318**	**0.802**	**0.437**
Qwen2.5-7B	base	0.779	**0.757**	0.856	2.8%	0.291	0.553	0.245	0.573	0.345
	instruct	**0.717**	0.785	**0.816**	14.6%	**0.364**	**0.474**	**0.307**	**0.735**	**0.439**
Qwen2.5-32B	base	0.726	**0.726**	0.830	12.9%	0.351	0.473	0.282	0.721	0.382
	instruct	**0.674**	0.759	**0.787**	22.7%	**0.418**	**0.437**	**0.333**	**0.840**	**0.472**
OLMo 3 7B	base	**0.794**	**0.761**	**0.850**	0.0%	**0.267**	**0.568**	**0.225**	0.470	0.256
	Instruct (RLVR)	0.823	0.851	0.884	−5.5%	0.257	0.604	0.220	**0.485**	**0.317**
OLMo 3.1 32B	base	0.746	**0.743**	0.835	9.1%	0.334	0.497	0.273	0.652	0.370
	Instruct (RLVR)	**0.701**	0.763	**0.812**	17.6%	**0.391**	**0.467**	**0.312**	**0.779**	**0.392**
Chance	patient-blind	0.794	0.774	—	0%	0.238	—	0.272	0.102	0.149
Oracle	reference	0.265	0.18–0.23	—	100%	0.735	0.000	—	0.969	0.975
	one answer: reference mode	—	0.694	—	—	—	—	—	—	—
	one answer: true pathology	—	0.714	—	—	—	—	—	—	—

Table 13: Stated vs. sampled readouts on DDXPlus (T=0.7, 980 cases, valid samples). *One list*: one sample's stated differential (per-case mean over valid samples). *Samples*: empirical distribution of top answers. *List w/o true*: TV after removing the true pathology from list and reference. *List TV skill*: the share of the gap between the patient-blind list (0%) and the reference's own top five (100%) that one list closes. *Cond. TV*: TV between the list and the reference restricted to the named conditions and renormalized (how the list allocates probability, independent of how much mass it names). DDF1: set F1 of the named conditions against the reference's support. *Recall@5*: the list contains the true pathology. *Severe named*: share of severe conditions with reference probability ≥ 0.1 that one list names. Chance: the top five of the average reference differential as a fixed list, and 32 draws from that average as samples. Oracle: the reference's own top five and reference draws; samplers always giving the reference mode or the true pathology. Bold: better model of each pair.

Appendix E. The stated readout: elicitation and robustness

E.1. Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 14). The results match the short list for Qwen. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038); OLMo-3-7B-Instruct's full distribution moves closer although its five-item list does not. Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.013, +0.127]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) and narrowly for Qwen2.5-7B-Instruct (+1.6), but not for OLMo-3-7B-Instruct (−9.5).

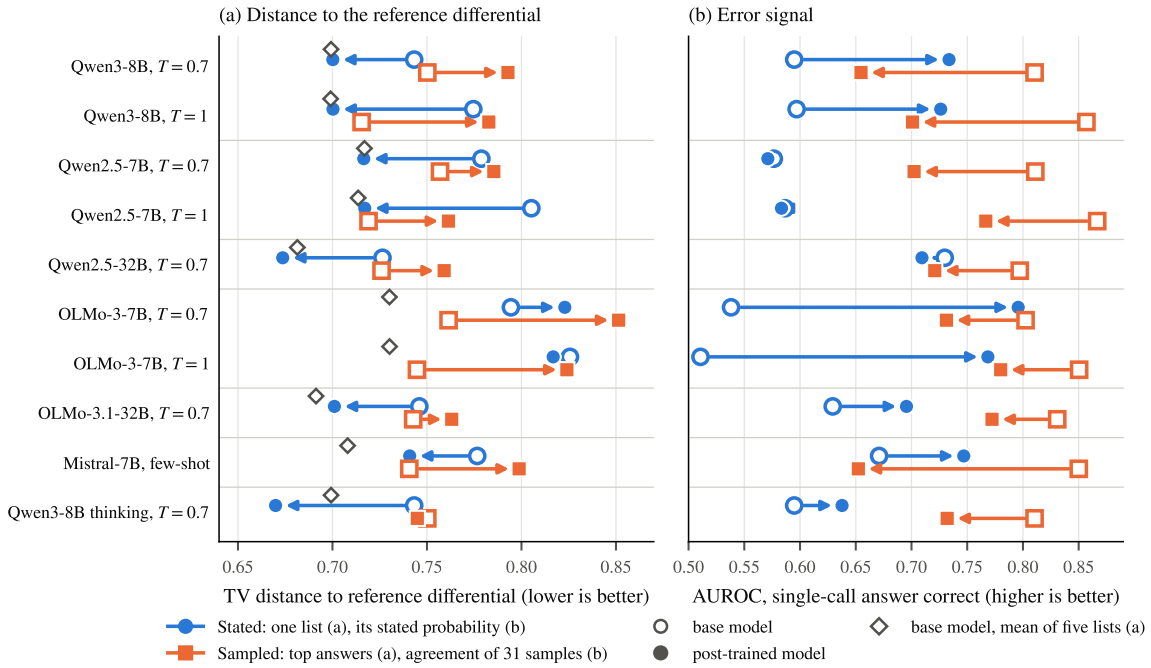


Figure 4: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; Mistral-7B with the few-shot prompt; the last row compares Qwen3-8B-Base with Qwen3-8B with thinking on). (a) TV to the reference differential of one stated list (blue circles) and of the sampled top answers (orange squares); grey diamonds: the mean of the base model's first five lists. (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

	TV to reference↓		AUROC, single-call answer correct			Top-3: one
Model	One distribution	Samples	Stated p	Agree@5	Agree@32	dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 14: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV to the reference of one sample's stated distribution and of the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

E.2. Breadth within one list and across calls

A five-item list cannot get shorter, so breadth within one call and breadth across calls need separate measures (Table 15). Within one call, post-trained lists name as many conditions as base-model lists (4.9–5.0) and carry more of the reference: 5–9 points more of its probability and 7–13 points more of the conditions it weights at least 0.1 (the upper ends from the later Qwen2.5-72B pair), significant for every pair but OLMo 3 7B, whose lists do not change. Across 32 calls the stated readout narrows: because post-trained lists vary little, their union names 4.6–10.0 fewer conditions than the union of the base model's lists, covers 10–13 points less reference mass and 7–14 points fewer warranted conditions (all significant). The union of 32 post-trained lists still covers more than twice the reference mass of the same model's 32 sampled answers (0.47–0.59 against 0.20–0.28). Narrowing across calls therefore affects both readouts, but much less the stated one, and within one call a list keeps its coverage (OLMo 3 7B's lists get worse under greedy decoding, Appendix F.3).

Sets of equal size. A five-item list could beat the sampled answers simply because it names more conditions. With sets capped at three, the top three of one list against the three most frequent of 32 sampled answers (zero-shot, T=0.7; Table 40), the samples contain the true pathology more often for every base model (by 7.3–13.6 points; Qwen2.5-72B 7.4), while for every post-trained Qwen model the list does (by 5.6–13.7 points; Qwen2.5-72B 8.4) and the OLMo Instruct models are about even (+1.9 and −0.6). A post-trained model's three most frequent answers are often fewer than three conditions, though; at exactly equal size (Appendix F.6) its two readouts are close (−0.5 to −5.2 points; the samples significantly ahead at 32B), while base models' samples still win by 8.5–14.6. The design therefore does not favor lists: it favors the samples for base models, and post-training removes that advantage by collapsing the samples, which is what a post-trained list's lead reflects. Comparing the full first list with all 32 samples, base models' samples miss the true pathology while their first list names it for only 2.9–3.5% of patients, against 7.2–19.9% for the post-trained models (Qwen3-8B 19.9, Qwen2.5-7B 14.7, Qwen2.5-72B 14.1, OLMo 3 7B 13.1, Qwen2.5-32B 11.0, OLMo 3.1 32B 7.2).

E.3. Free-length lists

The stated readout uses five-item lists, so a narrower view could hide behind the fixed length. We repeated the three Qwen pairs at T=0.7 with a prompt that asks for every condition the model considers plausible, as many or as few as it judges appropriate (Table 16). Length moves both ways: Qwen3-8B's lists grow after post-training (7.0 to 9.9 conditions), while the Qwen2.5 instruct models write shorter lists than their base models (5.2 against 10.7, and 4.0 against 5.8), whose lists often run on until the token limit. Whatever their length, post-trained lists contain the true pathology more often (by 8.8–16.3 points) and are closer to the reference (TV −0.047 to −0.072). They name as much reference mass or more (+14.9, −1.9 n.s., +2.8 points) and, for both Qwen2.5 models, allocate it more like the reference (conditional TV −0.144 and −0.078; Qwen3-8B +0.026, n.s. after correction). The sampled answers collapse as under the standard prompt (2.5–2.9 fewer distinct answers). The fixed length therefore does not hide a narrower stated view.

Pair	Model	One stated list			32 stated lists together			32 sampled top answers	
		Conditions	Mass	Warranted	Conditions	Mass	Warranted	Distinct	Mass
Qwen3-8B	base	4.89	0.34	0.45	16.6	0.61	0.71	3.77	0.32
	post-trained	4.99†	0.40	0.52	7.6	0.47	0.61	1.27	0.22
Qwen2.5-7B	base	4.98	0.30	0.38	22.0	0.67	0.75	4.91	0.33
	post-trained	4.89	0.36	0.48	12.0	0.56	0.68	1.82	0.24
Qwen2.5-32B	base	4.96	0.35	0.46	20.9	0.70	0.80	3.78	0.35
	post-trained	4.97†	0.42	0.56	11.6	0.59	0.73	1.68	0.28
OLMo 3 7B	base	4.77	0.26	0.34	18.0	0.62	0.74	4.93	0.35
	post-trained	4.93	0.26†	0.34†	13.4	0.50	0.60	2.37	0.20
OLMo 3.1 32B	base	4.86	0.34	0.45	18.7	0.68	0.80	4.32	0.35
	post-trained	4.99	0.39	0.51	11.7	0.58	0.71	2.15	0.28
Qwen2.5-72B*	base	4.98	0.32	0.43	18.0	0.65	0.75	3.01	0.30
	post-trained	4.97†	0.42	0.55	8.9	0.54	0.68	1.38	0.25

Table 15: Breadth within one call and across calls (DDXPlus, zero-shot, T=0.7, 980 cases). *Conditions*: conditions named (one list; or the union of the lists of up to 32 valid samples); *mass*: reference probability of the named conditions; *warranted*: share of the conditions with reference weight ≥ 0.1 that are named. Sampled top answers: distinct answers and the reference mass they cover (Cov@32). Base models with fewer than 32 valid samples are scored on the valid ones, which understates their breadth across calls. Only samples whose top answer is one of the 49 conditions are used, so one-list values can differ from Table 13 by up to 0.01. †: post-trained minus base not significant (condition-clustered 95% interval includes zero); every other difference is. *: added after the headline tests were fixed (§D.5).

E.4. Ruling out format, accuracy and floor effects

Table 18 tests simpler explanations of the stated-list result. For reference, one list's recall@5 rises after post-training by 13.6, 16.2 and 12.0 points for Qwen3-8B, Qwen2.5-7B and Qwen2.5-32B at T=0.7 and by 21.9 and 24.0 for the 7–8B Qwen models at T=1, by 12.7 for OLMo 3.1 32B, and for OLMo 3 7B by 1.5 (n.s.) at T=0.7 and 10.0 at T=1 (condition-clustered intervals exclude zero except where marked). *Format.* The general models' post-trained lists and base-model lists at T=0.7 are nearly all complete (82–99% name five on-list conditions with probabilities; Qwen base models at T=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.778 vs. 0.716). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong, the post-trained Qwen list is still closer to the reference (-0.013 to -0.041 and -0.030 to -0.090 in TV across both temperatures; Table 18 shows T=0.7). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 $\rightarrow$ 0.436) and raises the samples' (0.557 $\rightarrow$ 0.613). *True pathology removed.* With the true pathology removed from both list and reference and both renormalized, the post-trained Qwen and OLMo 3.1 32B lists are still closer to the reference (-0.018 to -0.056; n.s. for Qwen3-8B

				One stated list					Sampled top answers	
Pair	Model	Length	Contains true	Recall@5	TV↓	Mass	Cond. TV↓	Severe	Distinct	pass@32
Qwen3-8B	base	7.0	0.692	0.654	0.733	0.370	0.517	0.389	**3.86**	**0.735**
	post-trained	9.9	**0.855**	**0.799**	**0.660**	**0.519**	0.535	0.544	1.28	0.626
Qwen2.5-7B	base	10.7	0.659	0.552	0.765	0.401	0.623	0.444	**4.94**	0.684
	post-trained	5.2	**0.747**	**0.727**	**0.698**	0.383	**0.469**	0.446	2.03	0.627
Qwen2.5-32B	base	5.8	0.731	0.700	0.726	0.362	0.464	0.400	**4.26**	0.767
	post-trained	4.0	**0.839**	**0.835**	**0.679**	**0.389**	**0.380**	0.446	1.71	0.751

Table 16: Free-length lists (DDXPlus, zero-shot, T=0.7, 980 cases, 32 samples): the prompt asks for every condition the model considers plausible, as many or as few as it judges appropriate. *Length*: conditions per valid list; *contains true*: the list names the true pathology; *mass*: reference probability of the named conditions; *severe*: share of warranted severe conditions named. Bold: significantly better model of the pair (condition-clustered bootstrap, Holm across the table's 21 tests); length and severe named are not ranked. Differences in the text are paired over cases where both models give a valid list, so they can differ slightly from differences of the levels shown.

at T=0.7), and DDXPlus's set F1 against the reference's support improves by 0.040–0.083 (Table 46). TV, mass and DDF1 are not independent, since a list's TV is the reference mass it leaves out plus the mass it misallocates among the conditions it names, and the second term grows mechanically with the first. The conditional TV, between the list and the reference restricted to the named conditions and renormalized, removes this: it falls after post-training for every Qwen pair and OLMo 3.1 32B (−0.025 to −0.078, all surviving correction) and rises for OLMo 3 7B at T=1 (+0.064), so the Qwen lists both name better conditions and weight them more like the reference. *Patient-blind list.* A fixed list of the five conditions with the highest average reference probability scores TV 0.794, mass 0.238 and DDF1 0.272. Every model's list beats it in TV except OLMo 3 7B's and Qwen2.5-7B's base list at T=1; in DDF1 several base lists do not (Qwen2.5-7B 0.245, OLMo 3 7B 0.225), and the post-trained Qwen and OLMo 3.1 32B lists exceed it by 0.035–0.061. *Over-naming.* As a share of named conditions, post-trained lists name fewer that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55) and fewer unwarranted severe conditions (Qwen2.5-32B 0.80 → 0.73), but as a count per list the change is mixed: from 0.22 fewer unwarranted severe conditions (OLMo 3.1 32B) to 0.35 more (OLMo 3 7B). *Severe conditions one by one.* Table 17 tests each of the 17 severe conditions, counting it wherever the reference gives it any weight. After Benjamini–Hochberg correction, 2–7 conditions per zero-shot pair are named significantly less often after post-training and 0–6 significantly more often, and no pair names more conditions less often than a random swap of base and post-trained lists would. With the warranted threshold at 0.01 or 0.05 instead of 0.1, post-trained lists still name more severe conditions in seven of eight zero-shot settings (+2.5 to +8.1 points; OLMo 3.1 32B n.s.). That count test is weak against an overall upward shift, so Table 17 also gives the mean shift across the 17 conditions: it rises significantly in seven of eight zero-shot settings but falls with worked examples for Qwen2.5-32B (−0.012) and OLMo 3 7B (−0.018), and is flat for Mistral (−0.009, p=0.05). Significant losses include acute pulmonary edema

Pair (base → post)	Named less / more often	Significant less / more	P(as many less often)	Mean shift (17 conditions)	P	Largest loss (shift)
Qwen3-8B, T=0.7	5 / 12	2 / 3	0.98	+0.040	0.001	acute pulmonary edema (−0.14)
Qwen3-8B, T=1	6 / 11	2 / 4	0.94	+0.060	0.001	stable angina (−0.06)
Qwen2.5-7B, T=0.7	5 / 12	2 / 2	0.98	+0.029	0.001	stable angina (−0.16)
Qwen2.5-7B, T=1	4 / 13	2 / 5	0.99	+0.065	0.001	stable angina (−0.13)
Qwen2.5-32B, T=0.7	6 / 11	3 / 5	0.93	+0.047	0.001	acute pulmonary edema (−0.19)
OLMo 3 7B, T=0.7	8 / 9	7 / 6	0.68	+0.046	0.001	acute pulmonary edema (−0.24)
OLMo 3 7B, T=1	8 / 9	6 / 6	0.68	+0.061	0.001	NSTEMI/STEMI (−0.12)
OLMo 3.1 32B, T=0.7	7 / 10	5 / 0	0.82	+0.001	0.876	acute pulmonary edema (−0.16)
Qwen3-8B, few-shot	5 / 12	1 / 4	0.97	+0.061	0.001	acute pulmonary edema (−0.16)
Qwen2.5-7B, few-shot	8 / 9	4 / 2	0.69	+0.034	0.001	stable angina (−0.15)
Qwen2.5-32B, few-shot	10 / 7	5 / 3	0.29	−0.012	0.009	laryngospasm (−0.25)
OLMo 3 7B, few-shot	9 / 8	7 / 3	0.51	−0.018	0.018	croup (−0.51)
OLMo 3.1 32B, few-shot	9 / 8	4 / 2	0.50	+0.011	0.012	stable angina (−0.25)
Mistral-7B, few-shot	11 / 6	3 / 0	0.17	−0.009	0.052	pulmonary embolism (−0.31)

Table 17: The 17 severe conditions one by one (DDXPlus). For each condition, the share of a model's lists that name it, over the cases whose reference gives it any weight; post-trained minus base, with intervals that resample the true pathologies (50,000 resamples for the zero-shot rows, 10,000 for the few-shot rows) and Benjamini–Hochberg correction across the 17 conditions. P columns compare the count of conditions named less often and the mean shift across conditions with a null that swaps each case's base and post-trained lists at random (2,000 swaps); the count test is weak against an overall upward shift, so the mean shift and the largest single loss are shown too.

(Qwen3-8B 0.75 → 0.61, Qwen2.5-32B 0.63 → 0.44, OLMo 3 7B 0.56 → 0.31, OLMo 3.1 32B 0.57 → 0.41), stable angina (Qwen2.5-7B 0.25 → 0.08), NSTEMI/STEMI (OLMo 3 7B 0.31 → 0.14, OLMo 3.1 32B 0.49 → 0.36) and myocarditis (Qwen2.5-32B 0.14 → 0.05); significant gains include pulmonary embolism (Qwen2.5-7B 0.29 → 0.56, OLMo 3 7B 0.20 → 0.65), NSTEMI/STEMI (Qwen2.5-7B 0.32 → 0.52) and unstable angina (Qwen3-8B 0.14 → 0.37). Among single stages, OLMo 3 7B's RLVR step and OLMo 3.1 32B's SFT and RLVR steps do name more conditions less often than chance (12–13 of 17, p=0.03–0.04).

Sampling from the list. Replacing each sample's top answer with a draw from its own stated list raises pass@32 for every model but the OLMo 3 7B base (0.700 against 0.705). Compared like for like, post-trained lists reach higher than their base model's lists in five of seven comparisons (Qwen3-8B 0.812 vs. 0.787 at T=0.7, Qwen2.5-32B-Instruct 0.883 vs. 0.856, OLMo 3.1 32B Instruct 0.847 vs. 0.840), but not for Qwen3-8B at T=1 (0.815 vs. 0.834) or OLMo 3 7B (0.574 vs. 0.700), and a uniform guesser reaches 0.48. Drawn pass@32 stays below the share of cases whose lists' union contains the true pathology (Table 2; e.g. Qwen3-8B 0.812 against 0.847), so drawing realizes most, not all, of what the lists contain. Against the base model at its best prompt, with worked examples, drawn pass@32 is not significantly different for Qwen3-8B, Qwen2.5-7B and Qwen2.5-72B (+0.3, +3.2, +0.5 points; chance-corrected +2.1, −1.0, −4.5, all intervals include zero), though equivalence within 5 points is shown only for Qwen3-8B before chance correction (90% interval $[-4.4, 4.9]$), but, chance-corrected, short for Qwen2.5-32B and OLMo 3.1 32B (−1.3 and −4.1 raw; chance-corrected −7.8 $[-13.5, -1.9]$ and −7.6 $[-14.6, -0.4]$) and for OLMo 3 7B (−28.3

Model (T=0.7)	Complete lists	TV of one list↓ complete only	top-1 right	top-1 wrong	Excess TV over floor↓ One list	Samples	Severe named ref. < 0.1
Qwen3-8B-Base	0.90	0.742	0.659	0.794	0.479	**0.557**	0.815
Qwen3-8B	0.99	**0.700**	**0.646**	**0.765**	**0.436**	0.613	**0.712**
Qwen2.5-7B	0.96	0.778	0.686	0.815	0.514	**0.575**	0.847
Qwen2.5-7B-Instruct	0.91	**0.716**	**0.656**	**0.758**	**0.452**	0.605	**0.822**
Qwen2.5-32B	0.96	0.725	0.662	0.779	0.462	**0.543**	0.797
Qwen2.5-32B-Instruct	0.98	**0.673**	**0.638**	**0.706**	**0.409**	0.579	**0.729**
OLMo-3-1025-7B	0.82	**0.797**	0.682	**0.827**	**0.530**	**0.522**	**0.808**
OLMo-3-7B-Instruct	0.95	0.822	**0.662**	0.878	0.558	0.671	0.814
OLMo-3-1125-32B	0.89	0.743	0.664	0.811	0.481	**0.530**	0.773
OLMo-3.1-32B-Instruct	0.99	**0.701**	**0.646**	**0.754**	**0.436**	0.583	**0.731**

Table 18: Robustness of the stated-list result (DDXPlus, T=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the TV of the matching oracle (the reference's top five for one list; the expected TV of as many reference draws as the model has valid samples, for the samples). Severe named, ref. < 0.1: among lists naming at least one severe condition, the share of those conditions that the reference gives less than 0.1 (over-naming). Bold: better model of each pair. Excess TV is averaged over samples against each case's floor, so it is not exactly the difference of the mean TV and the mean floor. pass@32 with top answers drawn from the stated lists is in Table 2.

raw, −22.4 chance-corrected); drawn answers reach 0.09–0.13 against the wrong patient's diagnosis, the base model's worked-example samples 0.05–0.16. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

E.5. Can the model draw from its own list?

Drawing each sample's top answer from its own stated list recovers the zero-shot base model's pass@32 for every model except the OLMo 3 7B checkpoints (Table 2), but that draw happens outside the model. To test whether a post-trained model can do it itself, in the spirit of prompting for verbalized distributions (Zhang et al., 2026), we gave each of 32 calls its own random number u (stratified over $[0, 1]$) and asked the model to state its list as usual and then pick the condition at which the running total of its rescaled top-five probabilities first reaches u (Table 19). Picks are almost always present and on the model's own list, but the rule is followed only partly: in 39–40% of calls for the 7–8B Qwen models, 58% and 65% for Qwen2.5-32B-Instruct and OLMo-3-7B-Instruct, and 84% for OLMo-3.1-32B-Instruct. Four of five models recover breadth: 1.1–4.3 more distinct answers and 7.2–26.8 more points of pass@32 than their ordinary samples, which leaves none of them significantly below its base model's pass@32 and OLMo-3.1-32B-Instruct significantly above it (+5.7). Qwen3-8B follows the rule as often as Qwen2.5-7B-Instruct but rarely picks a lower-ranked condition when the rule calls for one: its picks are as narrow as its samples (1.25 distinct answers, against 5.96 if it followed the rule) and stay 15.7 points below its base model. Where

Model	Follows the rule	Distinct top answers seeded	ordinary	base	pass@32 seeded	− ordinary	− base	pass@1 − ordinary	pass@32 if rule followed	Agreement AUROC seeded	ordinary
Qwen3-8B	39%	1.25	1.27	3.79	0.574	−3.3	−15.7	−3.3	0.821	0.631	0.655
Qwen2.5-7B-Instruct	40%	2.94	1.82	4.94	0.670	+7.2	−2.6†	−2.1	0.792	0.735	0.702
Qwen2.5-32B-Instruct	58%	4.69	1.68	3.82	0.860	+11.4	+6.5†	−20.3	0.887	0.760	0.721
OLMo-3-7B-Instruct	65%	5.84	2.38	5.12	0.691	+26.8	−3.3†	+1.5†	0.697	0.785	0.732
OLMo-3.1-32B-Instruct	84%	6.44	2.15	4.39	0.856	+10.0	+5.7	−20.3	0.866	0.695	0.772

Table 19: Seeded pick (DDXPlus, zero-shot, T=0.7, 980 cases). Each of 32 calls receives its own random number u (stratified over $[0, 1]$) and is asked to state its list and then pick the condition at which the running total of its rescaled top-five probabilities first reaches u. *Follows the rule*: share of calls whose pick is that condition. *Ordinary*: the same model's zero-shot samples; *base*: its base model's. *pass@32 if rule followed*: the rule applied by us to each call's own list. Differences in points; †: not significant (condition-clustered bootstrap, Holm across the 40 tests of this analysis). Agreement for the seeded calls is the share of the other 31 picks equal to the first list's top answer. Ordinary and base columns use this analysis's parser (top answer = first listed item naming one of the 49 conditions), so they differ slightly from Table 2 (e.g. OLMo 3 7B base pass@32 0.724 against 0.705).

recovery works best it costs first-answer accuracy: pass@1 falls by 20.3 points for both 32B models, as expected when answers are drawn in proportion to the list rather than taken from its top. The extra instruction leaves the stated lists unchanged (recall@5 within 0.02 and TV within 0.01 of the ordinary lists, except OLMo-3-7B-Instruct's, which improve), and agreement among the picks is no better an error signal than ordinary agreement for any model. Applying the rule ourselves to each call's list would give pass@32 of 0.70–0.89. Breadth is thus recoverable from the stated list, reliably when the drawing is done outside the model.

E.6. The readout reversal and silent errors

Table 20 and Figure 1 compare, within each model, the first valid sample's stated list with the model's 32 sampled top answers. For base models the samples find the true pathology more often (by 4.5–28.6 points, all intervals excluding zero), for post-trained models the list does (by 2.1–19.6 points; n.s. for OLMo 3.1 32B), and post-training moves the difference by 17–41 points in every pair. Scored against the true pathology of another patient with a different pathology, one list reaches 0.06–0.09, and 32 samples 0.05–0.13 before post-training and 0.01–0.04 after, so chance favors the post-trained list; after subtracting it, base models' samples stay significantly ahead in all eight pairs, and the post-trained list stays ahead in seven, significantly for Qwen3-8B, Qwen2.5-7B-Instruct and MedGemma-4B-IT. At every budget k the union of k calls' lists (invalid calls name nothing; mean over 20 random orders) contains the true pathology more often than k samples, for every model, but base models need two to eight lists to match their own 32 samples, while post-trained models' first list already exceeds them. A *silent error* is a wrong single-call answer that every other valid sample repeats, so that agreement is maximal and cannot flag it. Post-

Pair	Model	First list recall@5	32 samples pass@32	List − samples raw	List − samples chance-corrected	Silent errors	Silent errors flagged by stated p
Qwen3-8B	base	0.687	0.732	−4.5	−6.9	4%	19%
	post-trained	0.803	0.607	+19.6	+12.4	61%	20%
Qwen2.5-7B	base	0.582	0.695	−11.3	−11.9	3%	25%
	post-trained	0.737	0.598	+13.9	+7.9	34%	16%
Qwen2.5-32B	base	0.718	0.795	−7.7	−10.3	6%	7%
	post-trained	0.839	0.746	+9.3	+2.7†	34%	15%
Qwen2.5-72B	base	0.698	0.755	−5.7	−10.0	9%	21%
	post-trained	0.864	0.731	+13.4	+6.2†	46%	24%
OLMo 3 7B	base	0.465	0.705	−24.0	−24.3	1%	20%
	post-trained	0.489	0.422	+6.6	+1.3†	23%	12%
OLMo 3.1 32B	base	0.645	0.796	−15.1	−16.3	3%	19%
	post-trained	0.778	0.756	+2.1†	−3.7†	19%	5%
Mistral-7B*	base	0.594	0.791	−19.7	−17.8	3%	19%
	post-trained	0.753	0.646	+10.7	+5.1†	60%	26%
MedGemma-4B*	base	0.499	0.785	−28.6	−21.6	1%	17%
	post-trained	0.761	0.635	+12.7	+6.6	52%	29%

Table 20: Readout reversal and silent errors (DDXPlus, T=0.7, 980 cases; zero-shot except *, worked examples). *First list*: the first valid sample's stated list contains the true pathology. *List − samples*: first list minus pass@32 of the same model, in points; *chance-corrected*: each term minus what it reaches against the true pathology of another case with a different pathology. *Silent errors*: share of wrong first answers with which every other valid sample agrees, so agreement cannot flag them. *Flagged by stated p*: share of those referred when the 20% of cases with the lowest stated probability are referred. †: condition-clustered 95% interval includes zero. Post-trained minus base, the list − samples difference rises by 17–41 points in every pair (all intervals exclude zero).

training raises their share from 1–9% to 19–61% of errors (by 6.6–25.1 points of all cases, all intervals excluding zero). Referring the 20% of cases with the lowest stated probability catches 5–29% of the silent errors, close to the 20% expected by chance (significantly below it for OLMo-3.1-32B-Instruct, 4.9%), and referral by agreement catches at least as many errors overall as referral by stated probability for every post-trained model but OLMo-3-7B-Instruct and MedGemma-4B-IT (stated minus agreement −12.9 to +2.7 points; significant for Qwen2.5-7B-Instruct and Qwen2.5-32B-Instruct).

Appendix F. Controls and further checks: prompt, temperature, decoding, chance, severity labels and further families

This section collects the controls for the main comparison. Table 21 shows, for each control, whether each main claim holds; the subsections give the numbers. Real-case evidence is in Appendix H.

Control	(a) View kept, variation removed	(b) Samples collapse	(c) Much of the first-answer gain is prompt format	(d) Stated list improves (Qwen 7–8B, 72B)	(e, f) Severe conditions; must-not-miss prompt	(g) Voting and agreement weaken
Worked examples, reference lists (§F.1)	holds	holds, larger (15–30 pts)	holds in 4 of 6 (not Qwen3-8B, Mistral)	holds for Qwen3-8B, Qwen2.5-7B; TV only for Qwen2.5-32B; weaker for OLMo 3.1 32B	partly: mean falls for Qwen2.5-32B, OLMo 3 7B	holds
Worked examples from another model (§F.2)	–	holds (13–30 pts)	–	holds for Qwen and Mistral, not OLMo	no excess losses (count test)	holds
Matched temperature (§F.2)	–	Qwen3-8B holds; Qwen2.5-7B-Instruct matches at T=1.3	–	holds	–	same split as (b)
Greedy decoding, matched prompts (§F.3)	–	–	holds in 4 of 6 pairs; not Qwen3-8B, nor Mistral with shared examples	holds for Qwen3-8B, Qwen2.5-7B (not Qwen2.5-32B recall); OLMo 3 7B farther	–	–
Temperature frontier (§F.3)	–	holds for OLMo 3 7B and both 32B pairs; no matched point for Qwen3-8B, Qwen2.5-7B	–	list unchanged across T	–	agreement below base at every T
Must-not-miss prompt (§F.3)	–	–	–	–	coverage +6–15 pts, partly by length	–
Published must-not-miss sets (§F.4)	–	–	–	–	mean up in 7 of 8; specific losses only OLMo, Mistral, few-shot Qwen2.5	–
Format, accuracy, floor, complete lists (§E.4)	–	–	–	holds	–	–
Other families' base models (Table 38)	holds across families; blurs within	–	–	–	–	–
Every other model's view (Table 33)	partly: own base closest for 3 of 5; as close to nearest post-trained model	–	–	–	–	–

Table 21: Which claims each control supports. "Holds": the control reproduces the claim; "partly": in some pairs or settings; "fails": it reverses or disappears; –: not tested. Claims are lettered as in Table 4, which lists the evidence claim by claim; (h) is not tabulated here; see Table 5.

Control	(a) View kept, variation removed	(b) Samples collapse	(c) Much of the first-answer gain is prompt format	(d) Stated list improves (Qwen 7–8B, 72B)	(e, f) Severe conditions; must-not-miss prompt	(g) Voting and agreement weaken
Chance-corrected pass@32 (§F.5)	–	holds with examples (6–18 pts at the best prompt; Qwen3-8B, Mistral n.s.); zero-shot 32B loss vanishes	–	–	–	–
Ties, list size, validity, prevalence (§F.6)	–	holds at natural prevalence	–	holds at natural prevalence	–	agreement weaker without ties; top-3 list advantage is list size
Concentration, misses, coverage, common cases (§F.7)	–	holds on common cases	–	lists do not concentrate (Qwen, OLMo 3.1 32B)	fewer per-case misses	holds on common cases and at 50% coverage
32B pairs (§D.5)	holds	n.s. zero-shot; holds with examples	holds (both pairs)	holds for Qwen; partly OLMo	holds for Qwen; OLMo n.s.	partly: agreement drop n.s. for OLMo
Mistral-7B, third family (§F.8)	holds	holds	fails with shared examples	holds, smallest gain	no mean gain; 3 of 17 lost significantly	holds
Thinking mode (§G.1)	holds	fails: pass@32 restored, still narrow	–	holds, closest list	–	fails: agreement better again
Rare-disease cases (§H.5)	–	–	–	–	–	holds; agreement stays the better signal
Published case reports (§H.3)	–	holds for the authors' alternatives (Qwen, OLMo 3.1 32B); pass@32 for 1 of 5	–	holds (recall +2 to +8; alternatives unchanged)	–	partly: voting gains small for all; stated p catches up with agreement
Closed-set case reports (§H.4)	–	holds in all 4 pairs (alternatives −22 to −27 pts)	–	holds (recall +4 to +8; alternatives +3 to +7)	–	partly: agreement falls in all 4; voting gains small for all; stated p better only for Qwen3-8B
Free-length lists (§E.3)	–	holds (2.5–2.9 fewer distinct)	–	holds (true pathology +9 to +16 pts; TV lower)	–	–
Seeded pick from own list (§E.5)	–	undone for 4 of 5 models; not Qwen3-8B	–	list unchanged (OLMo 3 7B improves)	–	agreement not restored
Qwen2.5-72B pair (§D.5)	–	n.s. zero-shot; holds with examples (−12.3 pts)	holds (+3.7, n.s., with examples)	holds (recall +15.4; TV −0.059; also with examples)	holds (severe +9.6 pts)	holds (voting +0.0; agreement 0.824 → 0.719)
One diagnosis per call (§D.3)	–	holds (1.2–1.4 against 3.5–3.8 distinct; pass@32 −8.4)	–	–	–	holds (agreement −0.22, −0.14; voting $\leq$ 1.4)
MedGemma-4B, medical (§F.9)	–	holds with examples (7.1 → 1.4 distinct; −15.0 pts), n.s. after chance correction	fails (+21.7 with shared examples)	holds (recall +27.1; TV −0.077)	holds (severe +15.0 pts)	holds (voting +10.7 → +0.8; stated p overtakes agreement)

Table 22: Which claims each control supports (continued from Table 21).

F.1. Few-shot elicitation

The main analyses give every model the same zero-shot chat prompt, which base models were not trained to follow; they give invalid answers on up to 47% of samples, which we count as wrong. We therefore re-ran the five final base/post-trained pairs at T=0.7 with three worked examples in the prompt: three fixed DDXPlus training patients with different true pathologies, each followed by the top five conditions of its reference differential, with probabilities, in the JSON format the instructions ask for. Base models see the instructions and examples as a plain document and complete the next one (generation stops at the end of the list); post-trained models see the same instructions and examples as earlier chat turns. Invalid samples fall below 0.5% for every model (Table 23).

The examples help base models most. Their pass@1 rises from 0.178 to 0.306 for OLMo-3-1025-7B, from 0.326 to 0.528 for OLMo-3-1125-32B and from 0.500 to 0.647 for Qwen2.5-32B, and their pass@32 reaches 0.76–0.90, while their stated lists barely change (TV 0.712–0.787, against 0.726–0.794 zero-shot). The losses in the samples therefore grow (Table 24): post-training costs 15.2–29.9 points of pass@32 in every pair, including both 32B pairs, whose zero-shot losses did not survive correction, and every post-trained model's Cov@32 (0.21–0.25) is below every base model's (0.33–0.45) and below a uniform guesser's.

For the three Qwen pairs the stated-list improvement persists and mostly grows: TV −0.057 to −0.087, −0.055 to −0.073 with the true pathology removed, and DDF1 +0.051 to +0.081, all surviving correction. OLMo 3 7B remains the exception (TV +0.021, n.s.; +0.024 without the true pathology). For OLMo 3.1 32B the improvement shrinks: the post-trained list still places more reference mass on the conditions it names (+0.047), but its TV (−0.022), TV without the true pathology (−0.006) and DDF1 (+0.009) no longer survive correction. At 32B the stated-list result is therefore robust for Qwen2.5-32B but depends on the prompt for OLMo 3.1 32B. Post-trained lists still name 5–10 of the 17 severe conditions less often than their base model's.

The consequences replicate in all five pairs. Majority voting over 32 samples gains base models 6.2–10.9 points and post-trained models 0.2–1.1. The three most frequent of 32 sampled answers beat the top three of one stated list by 8.2–22.0 points for base models, while for post-trained models the list is ahead by 0.2–16.7. Agreement with four more samples is the better error signal for every base model (AUROC 0.79–0.84, against 0.67–0.73 for the stated probability), and the stated probability for every post-trained model (0.69–0.77, against 0.56–0.68); against agreement across 31 samples the stated probability wins clearly only for Qwen3-8B and Qwen2.5-7B-Instruct (+0.108 and +0.130, clustered) and ties for the others. This check uses one temperature and one set of examples, and the examples show the reference's own lists, which may help every model match the reference.

F.2. Two further controls

Reference-free few-shot examples. The examples of §F.1 show each training patient's reference top five, and so also the target of the evaluation. We re-ran the six final pairs with the same three patients but with each example list replaced by Qwen2.5-32B-Instruct's own greedy answer (on-list, with the true pathology first for two patients and second for one). The conclusions do not change (Table 25). Post-training costs 12.7–29.7 points of pass@32 in every pair. The stated list moves closer to the reference for the three Qwen pairs and,

Model	Invalid (%)	pass@1	pass@32	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5	AUROC agree@32
Qwen3-8B-Base	0.1	0.403	**0.809**	**0.383**	0.768	0.860	0.245	**0.732**	+8.4	−16.0	0.722	0.828	0.862
Qwen3-8B	0.0	**0.601**	0.639	0.221	**0.682**	**0.805**	**0.327**	0.786	+0.2	+14.5	0.693	0.563	0.584
Qwen2.5-7B (base)	0.4	0.398	**0.761**	**0.333**	0.784	0.879	0.238	**0.756**	+6.2	−15.6	0.734	0.792	0.827
Qwen2.5-7B-Instruct	0.0	**0.450**	0.535	0.220	**0.702**	**0.806**	**0.309**	0.798	+0.6	+16.7	0.757	0.594	0.626
Qwen2.5-32B (base)	0.1	0.647	**0.896**	**0.343**	0.712	0.847	0.298	**0.715**	+7.1	−8.2	0.726	0.838	0.875
Qwen2.5-32B-Instruct	0.0	**0.663**	0.744	0.252	**0.654**	**0.775**	**0.349**	0.758	+0.6	+10.1	0.718	0.647	0.694
OLMo-3-1025-7B (base)	0.2	0.306	**0.857**	**0.454**	**0.787**	**0.859**	**0.245**	**0.712**	+10.9	−22.0	0.674	0.823	0.871
OLMo-3-7B-Instruct	0.1	**0.338**	0.558	0.213	0.808	0.884	0.238	0.830	+0.7	+0.2	0.733	0.668	0.696
OLMo-3-1125-32B (base)	0.4	0.528	**0.888**	**0.397**	0.719	0.818	0.302	**0.699**	+8.9	−13.4	0.670	0.826	0.865
OLMo-3.1-32B-Instruct	0.0	**0.575**	0.697	0.249	**0.697**	**0.812**	**0.311**	0.772	+1.1	+3.7	0.772	0.684	0.744

Table 23: Few-shot elicitation on DDXPlus (T=0.7, 980 cases, 32 samples; three worked examples in the prompt for every model, §F.1). Invalid: samples naming no listed condition. TV: distance to the reference differential, for one stated list (with the true pathology removed from both in "w/o true") and for the sampled top answers. Vote@32 gain and top-3 list minus top-3 of the samples are in accuracy points. AUROC: for the single-call answer being correct, of its stated probability and of agreement with the next 4 or the next 31 samples. Bold: better of the pair.

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Reference mass of one list
Qwen3-8B	−12.4 / −17.0	+0.043 / +0.054	−0.043 / −0.087	−0.018† / −0.055	+0.042 / +0.081	+0.062 / +0.113
Qwen2.5-7B	−9.7 / −22.7	+0.028† / +0.042	−0.062 / −0.082	−0.041 / −0.073	+0.062 / +0.072	+0.073 / +0.124
Qwen2.5-32B	−4.9† / −15.2	+0.033 / +0.044	−0.053 / −0.057	−0.043 / −0.072	+0.051 / +0.051	+0.067 / +0.084
OLMo 3 7B	−28.3 / −29.9	+0.090 / +0.118	+0.029† / +0.021†	+0.034 / +0.024	−0.006† / −0.007†	−0.010† / −0.002†
OLMo 3.1 32B	−4.0† / −19.1	+0.020† / +0.074	−0.045 / −0.022†	−0.023 / −0.006†	+0.040 / +0.009†	+0.056 / +0.047

Table 24: Base → post-trained differences under the zero-shot prompt / the few-shot prompt (DDXPlus, T=0.7). †: not significant under the condition-clustered bootstrap with Holm correction (over 112 tests for the zero-shot and 70 for the few-shot pairs; Appendix J).

now clearly, for Mistral (TV −0.053 to −0.077; −0.031 to −0.059 with the true pathology removed), but not for OLMo 3 7B or OLMo 3.1 32B, and wherever the list improves its allocation among the named conditions also moves closer to the reference (conditional TV −0.051 to −0.100). Voting helps base models by 3.7–13.7 points and post-trained ones by at most 1.4; the top three of one list beats the three most frequent samples for every post-trained model; the stated probability beats agreement as an error signal for every post-trained model and agreement beats it for every base model; and no pair names more severe conditions less often than chance. Five of six base models score lower at k=1 than with reference-derived examples (OLMo-3-1125-32B 0.453 against 0.528).

Matched temperature. Post-trained models are sharper than their base models at the same nominal temperature, so part of the narrowing could be colder effective sampling. We sampled Qwen3-8B and Qwen2.5-7B-Instruct at T=1.3 and 1.6 (Table 26), and the two models differ. Qwen3-8B barely broadens: at T=1.6 it gives 1.92 distinct answers and reaches 0.684 at pass@32, still below its base model at T=0.7 (0.732) and at T=1 (0.789;

Pair (base → post)	pass@32 (points)	TV samples	TV, one list all	TV, one list w/o true	DDF1	Cond. TV
Qwen3-8B	−14.6	+0.049	−0.077	−0.049	+0.079	−0.078
Qwen2.5-7B	−19.7	+0.031†	−0.077	−0.059	+0.074	−0.094
Qwen2.5-32B	−14.5	+0.036	−0.053	−0.057	+0.054	−0.051
OLMo 3 7B	−29.7	+0.116	−0.007†	+0.013†	−0.002†	−0.015†
OLMo 3.1 32B	−17.4	+0.073	−0.016†	+0.002†	+0.009†	−0.005†
Mistral-7B	−12.7	+0.038†	−0.059	−0.031	+0.057	−0.100

Table 25: Few-shot control with reference-free examples (DDXPlus, T=0.7, 980 cases): the same three training patients as in §F.1, but each example list is Qwen2.5-32B-Instruct's own greedy answer rather than the reference's top five. Post-trained minus base. †: not significant (condition-clustered bootstrap, Holm over 84 tests; conditional TV Holm-corrected as a separate family).

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV samples	TV one list	Vote@32 gain	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.7	11.3	0.388	0.732	3.77	0.750	0.743	+3.4	0.595	0.810
	1	22.3	0.282	0.789	6.60	0.715	0.775	+10.4	0.597	0.857
Qwen3-8B	0.7	0.0	0.559	0.607	1.27	0.793	0.700	+0.3	0.734	0.655
	1	0.0	0.556	0.643	1.53	0.783	0.700	+1.4	0.726	0.701
	1.3	0.0	0.552	0.668	1.73	0.776	0.701	+1.0	0.707	0.724
	1.6	0.0	0.549	0.684	1.92	0.770	0.701	+1.8	0.715	0.745
Qwen2.5-7B	0.7	0.9	0.352	0.695	4.91	0.757	0.779	+5.2	0.576	0.811
	1	18.4	0.239	0.766	8.23	0.719	0.805	+14.1	0.587	0.867
Qwen2.5-7B-Instruct	0.7	0.0	0.478	0.598	1.82	0.785	0.717	+0.8	0.571	0.702
	1	0.0	0.464	0.663	2.65	0.761	0.717	+4.4	0.583	0.767
	1.3	2.2	0.436	0.704	3.38	0.745	0.722	+5.6	0.602	0.794
	1.6	30.5	0.304	0.726	3.88	0.729	0.749	+10.0	0.637	0.814

Table 26: Matched-temperature control (DDXPlus, 980 cases, 32 samples, zero-shot prompt): the post-trained models sampled at higher temperatures (top-p 1) against their base models at T=0.7 and T=1. Columns as in Tables 2, 13 and 40.

−10.5 points, surviving correction); voting gains stay under 2 points, agreement stays weaker than for the base model, and its stated list does not change (TV 0.701). Qwen2.5-7B-Instruct does broaden: at T=1.3, with 2% invalid answers, it gives 3.38 distinct answers, matches its base model's pass@32 at T=0.7 (0.704 against 0.695), gains 5.6 points from voting, and agreement is again the better error signal (0.794 against 0.602), at a cost of 4.2 points of pass@1; at T=1.6, 31% of its answers are invalid. Its stated list stays closer to the reference than the base model's (0.722 against 0.779). For some models, then, much of the sampled narrowing is sharper sampling that a higher temperature undoes; the stated-list result is not.

F.3. Greedy decoding, temperature frontier and a must-not-miss prompt

Greedy single call. Deployed systems often decode greedily. We compared one greedy answer per case under three prompt pairings (Table 28): both arms with the reference-derived worked examples of §F.1 (*same examples*); base models with the reference-free examples of Table 25 against zero-shot post-trained models, so that neither arm sees a reference list (*reference-free base*); and reference-example base models against zero-shot post-trained models, the pairing we first ran. Under every pairing post-training significantly raises top-1 in at most two of six pairs. With the same examples these are Qwen3-8B (+13.7) and Mistral (+10.8); Qwen2.5-7B gains +1.8 and the other three lose 4.8–5.8 points, all n.s. With reference-free base models only Qwen3-8B gains (+10.7), while Qwen2.5-32B (−8.0) and OLMo 3 7B (−14.5) lose significantly. The example lists matter little to the base models (top-1 within 2.7 points, except OLMo 3.1 32B, 0.630 against 0.559) but raise five of the six post-trained models (Mistral 0.417 to 0.577, OLMo 3 7B 0.258 to 0.345; Qwen2.5-7B-Instruct falls to 0.458). Under every pairing the two smaller Qwen models' lists contain the true pathology more often (recall@5 +16.8 to +26.0) and name more warranted severe conditions, every Qwen list is closer to the reference, and OLMo 3 7B's lists are farther. Together with the few-shot comparison, where the sampled pass@1 gain is already only 1.6–5.2 points in four of six pairs, much of post-training's advantage at one sampled call for the 7–8B Qwen models (+12.6 to +27.5 points at T=0.7 and 1, §4.1) is therefore prompt format and the base model's sampling noise; Qwen3-8B and Mistral keep a real gain. These are not equivalence results: with the margin of 5 points fixed in advance, no pair's greedy top-1 is equivalent to its base model's under any pairing (TOST at the 5% level), so the gain is no longer significant rather than shown to vanish.

Temperature frontier. With worked examples and top-p fixed at 1, we sampled the three 7–8B pairs at T = 0.4, 0.7, 1.0 and 1.3 (Table 29). Post-trained pass@1 barely moves with temperature (Qwen3-8B 0.60, Qwen2.5-7B-Instruct 0.43–0.46, OLMo-3-7B-Instruct 0.29–0.35), pass@32 rises (to 0.70, 0.66 and 0.74 at T=1.3), and the stated list barely changes (TV 0.68, 0.70 and 0.81–0.83), consistent with the concentration described by Yang et al. (2026). Base models trade pass@1 (Qwen3-8B-Base 0.45 to 0.18) for pass@32, which peaks at T=1 or 1.3 (0.85–0.88), and 25–36% of their answers are invalid at T=1.3. The frontiers differ by family. For Qwen3-8B the post-trained model is better at pass@1 and the base model at pass@32 at every temperature tested; for Qwen2.5-7B the curves meet (at pass@1 about 0.43, the base model at T=0.4 and the Instruct model at T=1.3 both reach pass@32 0.66); and for OLMo 3 7B the base model at T=0.4 beats its Instruct model at every temperature on both (0.368 and 0.803, against at best 0.345 and 0.737). Interpolating the base model's frontier at the Instruct model's pass@1, OLMo-3-7B-Instruct reaches 13.0–31.9 points less pass@32 at each temperature (condition-clustered intervals exclude zero); for Qwen3-8B and Qwen2.5-7B-Instruct the post-trained pass@1 lies above every base point, so no matched comparison exists. We repeated the frontier for the two 32B pairs (Table 27). There the base model at T=0.4 beats its post-trained model on both pass@1 and pass@32 at every temperature, and at the post-trained model's pass@1 the base frontier reaches 7.4–15.4 points more pass@32 for Qwen2.5-32B and 4.5–17.0 for OLMo 3.1 32B (intervals exclude zero except OLMo at T=1.3), although the zero-shot 32B losses do not survive

Pair	Model	pass@1 / pass@32 at T 0.4	0.7	1.0	1.3
Qwen2.5-32B	base	**0.686** / **0.859**	0.627 / 0.907	0.526 / 0.939	0.365 / 0.947
	Instruct	0.666 / 0.722	0.662 / 0.765	0.658 / 0.790	0.650 / 0.814
	matched difference	−15.4	−11.4	−9.3	−7.4
OLMo 3.1 32B	base	**0.590** / **0.829**	0.510 / 0.896	0.407 / 0.925	0.252 / 0.899
	Instruct	0.573 / 0.674	0.573 / 0.730	0.570 / 0.780	0.558 / 0.810
	matched difference	−17.0	−11.3	−6.6	−4.5†

Table 27: Temperature frontier for the 32B pairs (DDXPlus, three worked examples, top-p 1, 980 cases, 32 samples). Bold: the base model at T=0.4, which beats its post-trained model on both readouts at every temperature. *Matched difference*: post-trained pass@32 minus the base model's pass@32 interpolated at the post-trained model's pass@1, in points; †: condition-clustered 95% interval includes zero.

correction (§D.5). Agreement becomes a better error signal for post-trained models as temperature rises (Qwen3-8B 0.57 to 0.68) but stays below the base models' (0.81–0.92).

Must-not-miss prompt. The standard prompt asks for the most likely conditions. We asked the six post-trained models for the five most likely conditions and then up to three dangerous conditions to rule out (Table 30). Whole lists then name 0.44–0.58 of the severe conditions that the reference weights at least 0.1, 6–15 points more than with the standard prompt and 11–21 more than the base models' lists, and acute pulmonary edema is recovered for every model but Qwen3-8B (e.g. Qwen2.5-32B 0.44 → 0.63, OLMo 3 7B 0.31 → 0.58). The cost is 1.1–2.1 more unwarranted severe conditions per list, mostly pulmonary embolism, named in 88–100% of eligible cases, and for Qwen3-8B, Qwen2.5-7B-Instruct and Mistral a shorter most-likely part that contains the true pathology 5–10 points less often. Even when asked, no model names more than 0.58 of the warranted severe conditions.

F.4. Must-not-miss conditions defined outside DDXPlus

The severity label used above is the dataset's own. We therefore re-ran every severe-condition analysis with two published sets of conditions that must not be missed, mapped onto DDXPlus's 49 (Table 31): a harm-based set, the DDXPlus conditions among the 15 diseases that cause about half of serious misdiagnosis-related harms in the USA (Newman-Toker et al., 2024) (NSTEMI/STEMI, pulmonary embolism, pneumonia and pulmonary neoplasm, of which DDXPlus labels only the first two severe), and the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline (Gulati et al., 2021) (acute coronary syndrome as NSTEMI/STEMI and unstable angina, pulmonary embolism, esophageal rupture as Boerhaave, and pneumothorax). Their union has seven conditions. Neither set involves a judgment of these vignettes, and both were fixed before the analysis. On average the result is stronger than with DDXPlus's labels. Zero-shot, post-trained lists name 9–24 points more of the warranted union conditions in seven of eight settings (OLMo 3.1 32B −0.3, n.s.), and with worked examples 7–22 points more for the three Qwen models and

Pair	Top-1 accuracy: base, examples: reference	base, examples: ref.-free	post-trained: zero-shot	post-trained: examples	Top-1, post − base (points): same examples	ref.-free base	original
Qwen3-8B	0.471	0.450	0.558	0.607	+13.7	+10.7	+8.7†
Qwen2.5-7B	0.437	0.420	0.496	0.458	+1.8†	+7.6†	+5.7†
Qwen2.5-32B	0.712	0.686	0.606	0.664	−4.8†	−8.0	−10.6
OLMo 3 7B	0.403	0.403	0.258	0.345	−5.8†	−14.5	−14.5
OLMo 3.1 32B	0.630	0.559	0.558	0.571	−5.7†	−0.1†	−7.1†
Mistral-7B	0.468	0.446	0.417	0.577	+10.8	−2.9†	−5.1†

Pair	Recall@5, post − base (points): same	ref.-free	original	TV one list, post − base: same	ref.-free	original	Severe named, post − base: same	ref.-free	original
Qwen3-8B	+26.0	+21.6	+24.2	−0.108	−0.077	−0.088	+0.127	+0.127	+0.127
Qwen2.5-7B	+20.1	+18.3	+16.8	−0.074	−0.075	−0.060	+0.091	+0.119	+0.110
Qwen2.5-32B	+0.3†	−2.4†	−5.3†	−0.054	−0.035	−0.031	+0.058	+0.046	+0.062
OLMo 3 7B	+2.0†	−17.2	−14.1	+0.060	+0.067	+0.083	+0.004†	−0.058†	−0.020†
OLMo 3.1 32B	+2.5†	−0.5†	−2.0†	+0.004†	+0.004†	+0.012†	−0.005†	−0.033†	−0.042†
Mistral-7B	+16.7	+2.9†	+4.1†	−0.009†	+0.008†	+0.029	−0.034†	−0.034†	−0.059†

Table 28: Greedy single call (DDXPlus, 980 cases, temperature 0, one answer per case). Base models are prompted with the worked examples of §F.1, whose lists are the reference top five ("reference") or Qwen2.5-32B-Instruct's answers ("ref.-free"); post-trained models are prompted zero-shot or with the reference examples. Three comparisons: *same examples* (both arms see the reference examples), *ref.-free base* (reference-free base against zero-shot post-trained, so neither arm sees a reference list), and *original* (reference-example base against zero-shot post-trained). Differences are over cases where both models gave a valid answer, so they can differ slightly from the difference of the level columns. †: the condition-clustered 95% interval includes zero.

OLMo 3 7B (OLMo 3.1 32B +6.5 and Mistral +4.0, n.s.); the harm-based and chest-pain sets give the same zero-shot pattern. The losses of specific conditions, however, are confined to OLMo, Mistral and the few-shot Qwen2.5 models. No Qwen model names any union condition significantly less often zero-shot, while OLMo 3 7B loses NSTEMI/STEMI and unstable angina at both temperatures (and pulmonary neoplasm at T=1), OLMo 3.1 32B loses NSTEMI/STEMI, pulmonary embolism and pneumothorax, and with worked examples Mistral loses four of the seven (pulmonary embolism 31 points), Qwen2.5-7B-Instruct two and Qwen2.5-32B-Instruct one (Boerhaave). When the true pathology is itself in the union set (140 cases), zero-shot post-trained lists contain it among their top five 10–26 points more often for four of five pairs (OLMo 3 7B +9.9, n.s.). Scored against the union set, the must-not-miss prompt names 15–26 points more of the warranted conditions than the standard prompt and 8–20 points more than two combined standard lists, a larger share of its gain than under DDXPlus's labels, at a cost of 0.7–1.5 more unwarranted union conditions per list.

Model	T	Invalid (%)	pass@1	pass@32	Distinct	TV one list	AUROC stated	AUROC agree@32
Qwen3-8B-Base	0.4	0.1	0.447	0.727	3.64	0.772	0.687	0.806
	0.7	0.3	0.388	0.832	7.09	0.774	0.708	0.876
	1.0	3.0	0.304	0.880	10.33	0.800	0.703	0.909
	1.3	29.5	0.179	0.857	10.39	0.850	0.630	0.921
Qwen3-8B	0.4	0.0	0.602	0.632	1.14	0.681	0.693	0.569
	0.7	0.0	0.601	0.655	1.25	0.682	0.693	0.613
	1.0	0.0	0.599	0.676	1.40	0.682	0.695	0.651
	1.3	0.0	0.596	0.703	1.57	0.682	0.694	0.679
Qwen2.5-7B	0.4	0.4	0.421	0.662	3.13	0.778	0.748	0.805
	0.7	0.6	0.388	0.784	5.72	0.790	0.719	0.843
	1.0	2.6	0.333	0.841	8.53	0.815	0.692	0.899
	1.3	25.3	0.220	0.848	9.24	0.857	0.615	0.916
Qwen2.5-7B-Instruct	0.4	0.0	0.455	0.516	1.40	0.702	0.750	0.613
	0.7	0.0	0.448	0.556	1.78	0.703	0.747	0.643
	1.0	0.0	0.441	0.597	2.28	0.704	0.741	0.687
	1.3	0.2	0.430	0.660	2.98	0.707	0.720	0.729
OLMo-3-1025-7B	0.4	0.1	0.368	0.803	5.46	0.762	0.667	0.815
	0.7	0.4	0.294	0.865	9.62	0.795	0.663	0.879
	1.0	2.8	0.214	0.881	12.40	0.834	0.618	0.899
	1.3	36.2	0.110	0.801	10.69	0.888	0.552	0.882
OLMo-3-7B-Instruct	0.4	0.1	0.345	0.504	1.99	0.809	0.737	0.663
	0.7	0.1	0.335	0.602	2.95	0.808	0.740	0.716
	1.0	0.1	0.324	0.668	4.09	0.809	0.744	0.760
	1.3	5.6	0.286	0.737	5.88	0.831	0.680	0.775

Table 29: Temperature frontier with the few-shot prompt (DDXPlus, 980 cases, 32 samples, top-p 1 at every temperature).

F.5. Chance-corrected pass@32 and other models' views

Chance. With 49 listed conditions, part of pass@32 comes from answers that do not depend on the patient. We therefore also score each case's samples against the true pathology of a random case with a different pathology (Table 32). Base models' samples then still reach 0.05–0.19 at pass@32 and post-trained ones 0.01–0.04, far below the 0.48 of a uniform guesser, because both concentrate on conditions that fit the presentation. Subtracting these values shrinks the loss: with each model at its best prompt it is 5.9–17.6 points, significant in four of six pairs (not for Qwen3-8B, 7.7, or Mistral, 5.9); with the same reference examples for both, 7.2–17.6 points (Qwen3-8B n.s.); with another model's examples, 4.9–14.8 (Qwen3-8B and Mistral n.s.). Zero-shot, the corrected loss is 7.6 points for Qwen3-8B, 3.6 (n.s.) for Qwen2.5-7B and 23.7 for OLMo 3 7B, and it disappears at 32B (0.7 and −0.4 points, n.s.). For Qwen2.5-72B, added later, the corrected loss is 8.8 points at the best prompt (significant) and −0.9 zero-shot (n.s.).

Coverage corrected for chance. Cov@32 rewards spread: a uniform guesser covers 0.47 of the reference mass, more than any model. We therefore subtract, for each case,

Model	Valid (%)	Conditions likely + mnm	Recall@5 default	Recall@5 likely part	Severe named (ref. ≥ 0.1) base	Severe named (ref. ≥ 0.1) default	Severe named (ref. ≥ 0.1) mnm	Unwarranted severe default	Unwarranted severe mnm	Acute pulm. edema default	Acute pulm. edema mnm
Qwen3-8B	100	4.1 + 2.3	0.802	0.715	0.318	0.437	0.525	1.06	2.11	0.61	0.60
Qwen2.5-7B-Instruct	99	3.9 + 2.9	0.735	0.640	0.304	0.439	0.503	1.57	2.95	0.63	0.71
Qwen2.5-32B-Instruct	100	4.8 + 2.9	0.840	0.838	0.414	0.472	0.579	0.91	2.27	0.44	0.63
OLMo-3-7B-Instruct	96	3.7 + 2.5	0.485	0.501	0.332	0.317	0.444	1.15	2.60	0.31	0.58
OLMo-3.1-32B-Instruct	100	4.9 + 3.0	0.779	0.790	0.376	0.392	0.545	0.74	2.85	0.41	0.59
Mistral-7B-Instruct	87	3.7 + 2.7	0.613	0.562	0.352	0.329	0.473	0.86	2.29	0.59	0.68

Table 30: Must-not-miss prompt (DDXPlus, T=0.7, 980 cases, 32 samples): the five most likely conditions, then up to three dangerous conditions to rule out. *Default*: the same model with the standard zero-shot prompt; *base*: its base model with the few-shot prompt. Recall@5 of the most-likely part; severe conditions and unwarranted severe conditions (severe, reference < 0.1, per list) over the whole list; acute pulmonary edema: share of lists naming it where the reference gives it any weight.

the mass the same sampled answers cover in another patient's reference (a case with a different pathology; 20 permutations), which a patient-blind guesser scores at about zero, and do the same for the reference mass one list names (zero-shot, T=0.7). Corrected, post-training still lowers the samples' coverage in five of six pairs, by 2.0–9.1 points instead of 6.4–15.6 (Qwen3-8B −4.8 [−7.6, −2.3], Qwen2.5-7B −3.1, Qwen2.5-32B −2.7, OLMo 3 7B −9.1, OLMo 3.1 32B −2.0; Qwen2.5-72B −1.7 [−3.4, +0.2]), while one list's gain is almost unchanged (5.3–7.5 points, against 5.6–8.5 uncorrected; OLMo 3 7B +0.2, n.s.). Both readouts' changes therefore hold for patient-specific coverage.

Other models' views. The packing test of Table 38 compares a post-trained list with its base model's mean for the same patient against the base model's means for other patients with the same diagnosis, a lenient baseline that any list responding to the patient passes. Against other families' base models alone, every zero-shot post-trained list is closer to its own base model's mean (−0.031 to −0.138). Table 33 compares it with every other final model's mean for the same patient. For Qwen3-8B, Qwen2.5-32B and OLMo 3.1 32B the post-trained list is closer to its own base model's view than to any other base model's (by 0.036–0.058) and far closer than to the average post-trained model's (0.067–0.090), but about as close to the nearest other post-trained model's (n.s.). Qwen2.5-7B-Instruct is as close to Qwen3-8B-Base's view as to its own base model's, and OLMo-3-7B-Instruct is closer to Qwen3-8B-Base's and Qwen3-8B's. For the stronger models, then, a post-trained list stays close to its own base model's view, while post-trained lists also converge on one another.

F.6. Ties, list size, validity, prevalence and the reference ceiling

These checks use the zero-shot T=0.7 final pairs. *Ties.* Post-trained models' 32 samples are unanimous in 34–57% of cases (Qwen3-8B 78%), where agreement cannot rank anything. On the remaining cases agreement's AUROC is 0.61–0.66 for the post-trained models against 0.78–0.83 for their base models, so agreement weakens even where it is not tied. On those cases Qwen3-8B's stated probability no longer beats agreement (0.611 against 0.617): its advantage comes from cases where all samples agree, whereas OLMo-3-7B-Instruct's stated

Pair	Union set (7 conditions) base	post	post − base	sig. lost / gained	Harm-based post − base	Chest pain post − base
Qwen3-8B, T=0.7	0.494	0.584	+0.089	0 / 2	+0.053	+0.115
Qwen3-8B, T=1	0.425	0.585	+0.160	0 / 4	+0.135	+0.167
Qwen2.5-7B, T=0.7	0.315	0.536	+0.221	0 / 4	+0.267	+0.149
Qwen2.5-7B, T=1	0.303	0.539	+0.236	0 / 4	+0.287	+0.180
Qwen2.5-32B, T=0.7	0.432	0.595	+0.162	0 / 4	+0.164	+0.104
OLMo 3 7B, T=0.7	0.268	0.413	+0.144	2 / 3	+0.133	+0.101
OLMo 3 7B, T=1	0.275	0.412	+0.136	3 / 3	+0.135	+0.102
OLMo 3.1 32B, T=0.7	0.462	0.459	−0.003†	3 / 0	−0.033†	−0.020†
Qwen3-8B, few-shot	0.299	0.491	+0.191	0 / 3	+0.183	+0.093
Qwen2.5-7B, few-shot	0.239	0.455	+0.216	2 / 3	+0.249	+0.152
Qwen2.5-32B, few-shot	0.361	0.542	+0.180	1 / 3	+0.210	+0.096
OLMo 3 7B, few-shot	0.335	0.409	+0.074	2 / 3	−0.015†	+0.036†
OLMo 3.1 32B, few-shot	0.412	0.477	+0.065†	2 / 1	+0.080	−0.017†
Mistral-7B, few-shot	0.401	0.441	+0.040†	4 / 1	+0.024†	−0.071

Post-trained model	Warranted must-not-miss conditions named (union set) standard	two lists	must-not-miss	− standard	Unwarranted per list standard	two lists	must-not-miss
Qwen3-8B	0.584	0.612	0.775	+0.192	1.08	1.20	2.10
Qwen2.5-7B-Instruct	0.536	0.600	0.714	+0.178	0.94	1.14	2.33
Qwen2.5-32B-Instruct	0.595	0.648	0.814	+0.219	1.00	1.21	2.47
OLMo-3-7B-Instruct	0.413	0.479	0.564	+0.151	0.93	1.17	1.60
OLMo-3.1-32B-Instruct	0.459	0.527	0.715	+0.256	0.58	0.77	1.77
Mistral-7B-Instruct	0.358	0.417	0.615	+0.257	0.69	0.87	1.65

Table 31: Must-not-miss conditions defined outside DDXPlus (DDXPlus, 980 cases). *Harm-based*: the DDXPlus conditions among the 15 diseases behind about half of serious misdiagnosis-related harms in the USA (Newman-Toker et al., 2024) (NSTEMI/STEMI, pulmonary embolism, pneumonia, pulmonary neoplasm). *Chest pain*: the life-threatening causes of acute chest pain in the 2021 AHA/ACC guideline (Gulati et al., 2021) (NSTEMI/STEMI, unstable angina, pulmonary embolism, Boerhaave, spontaneous pneumothorax). *Union*: either set. Top: share of the set's conditions with reference probability ≥ 0.1 that one list names, and the number of the union's conditions named significantly less / more often (Benjamini–Hochberg over the 7 conditions). Bottom: the must-not-miss prompt (§F.3) scored against the union set; *two lists*: the union of two standard lists, a length-matched comparison; unwarranted: the set's conditions named that the reference weights below 0.1. †: the condition-clustered 95% interval includes zero.

probability stays ahead (0.729 against 0.605). *List size.* The three most frequent of 32 samples can be fewer than three answers. With $m = \min(3, \text{distinct sampled answers})$, the first m items of one list contain the true pathology about as often as the m most frequent samples for the post-trained models (−0.5 to −5.2 points; significantly lower for Qwen2.5-32B-Instruct and OLMo-3.1-32B-Instruct), while for base models the samples win by 8.5–14.6. The advantage of a post-trained list's top three is that it names three conditions where its samples offer fewer. *Validity.* Counting invalid samples as wrong (as everywhere in the paper) or scoring valid samples only leaves pass@32 unchanged and moves pass@1, mostly

Pair	Zero-shot raw	Zero-shot corrected	Reference examples raw	Reference examples corrected	Another model's examples raw	Another model's examples corrected	Each at its best prompt raw	Each at its best prompt corrected	Best prompt (base / post)
Qwen3-8B	−12.4	−7.6	−17.0	−7.7†	−14.6	−8.3†	−17.0	−7.7†	reference / reference
Qwen2.5-7B	−9.7	−3.6†	−22.7	−16.6	−19.7	−14.2	−16.3	−10.6	reference / zero-shot
Qwen2.5-32B	−4.9†	−0.7†	−15.2	−11.3	−14.5	−10.8	−15.0	−11.8	reference / zero-shot
OLMo 3 7B	−28.3	−23.7	−29.9	−17.6	−29.7	−14.8	−29.9	−17.6	reference / reference
OLMo 3.1 32B	−4.0	+0.4†	−19.1	−12.6	−17.4	−10.0	−13.2	−7.6	reference / zero-shot
Mistral-7B	—	—	−14.5	−7.2	−12.7	−4.9†	−13.2	−5.9†	reference / another
Qwen2.5-72B*	−2.4†	+0.9†	−12.3	−8.8	—	—	−12.3	−8.8	reference / reference
MedGemma-4B*	—	—	−15.0	−3.7†	—	—	−5.4†	+5.6†	reference / zero-shot

Table 32: Chance-corrected pass@32 (DDXPlus, T=0.7, 980 cases): post-trained minus base, in points. *Raw*: pass@32; *corrected*: pass@32 minus the pass@32 reached when each case's samples are scored against the true pathology of a random case with a different pathology (mean of 20 permutations), the part that does not depend on the patient (base models 0.05–0.19, post-trained 0.01–0.04). *Each at its best prompt*: each model at the prompt where its own pass@32 is highest. Top answers and validity as in every other table. †: condition-clustered 95% interval includes zero. Mistral's and MedGemma's base models do not answer the zero-shot prompt (§F.8, F.9). *: added after the headline tests were fixed; run zero-shot and with reference examples only (§D.5, F.9).

Post-trained model of	TV to own base mean	Closest other base model	own − it	Closest other post-trained model	own − it	Own − mean of other post-trained models
Qwen3-8B	0.368	OLMo-3-1125-32B	−0.058	Qwen2.5-32B-Instruct	+0.000†	−0.067
Qwen2.5-7B	0.464	Qwen3-8B-Base	+0.014†	Qwen2.5-32B-Instruct	+0.028	−0.041
Qwen2.5-32B	0.360	Qwen3-8B-Base	−0.036	OLMo-3.1-32B-Instruct	−0.018†	−0.090
OLMo 3 7B	0.628	Qwen3-8B-Base	+0.031	Qwen3-8B	+0.041	+0.002†
OLMo 3.1 32B	0.386	Qwen2.5-32B	−0.051	Qwen2.5-32B-Instruct	−0.006†	−0.079

Table 33: A stricter packing test (DDXPlus, zero-shot, T=0.7, 980 cases). For each post-trained model, the TV between its lists and the mean list of each other final model (the five base and the four other post-trained models) for the same patient. Negative differences: the post-trained list is closer to its own base model's view than to the comparison. †: the condition-clustered 95% interval includes zero.

for OLMo 3 7B, whose single-answer gain is +7.1 points with invalid samples counted as wrong and −3.5 (n.s.) over valid samples only, weighted to the natural prevalence below. *Prevalence.* The test sample is class-balanced (20 cases per pathology). Reweighted to the pathology frequencies of the full DDXPlus test split (134,529 patients), post-training costs 4.9–31.5 points of pass@32 (all intervals exclude zero, including both 32B pairs), raises recall@5 by 10.3–17.7 points for the Qwen models and OLMo 3.1 32B (−0.3 for OLMo 3 7B) and lowers one list's TV by 0.047–0.069 (OLMo 3 7B +0.031). *Reference ceiling.* The reference's own top five contains the true pathology in 96.9% of cases (97.3% at natural prevalence), against 0.47–0.84 for one model list. *What replaces acute pulmonary edema.* On the 54 cases whose reference gives acute pulmonary edema at least 0.1, the conditions whose share of lists rises most after post-training are mostly other must-not-miss conditions (pulmonary embolism for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct, +0.29 and

Pair	Entropy of one list (nats)	Lists that miss (share) severe true dx	Lists that miss (share) all warranted severe	Accuracy at 50% coverage, stated p / agreement base	Accuracy at 50% coverage, stated p / agreement post-trained	Agreement AUROC, both non-unanimous (n)
Qwen3-8B, zero-shot	1.29 → 1.16	0.38 → 0.20	0.50 → 0.39	0.52 / 0.69	0.74 / 0.68	0.74 → 0.62 (214)
Qwen2.5-7B, zero-shot	1.41 → 1.35	0.39 → 0.28	0.49 → 0.39	0.41 / 0.60	0.54 / 0.65	0.72 → 0.61 (477)
Qwen2.5-32B, zero-shot	1.32 → 1.40	0.31 → 0.16	0.46 → 0.35	0.68 / 0.75	0.74 / 0.77	0.77 → 0.64 (417)
OLMo 3 7B, zero-shot	1.41 → 0.89	0.55 → 0.46	0.59 → 0.50	0.31 / 0.46	0.44 / 0.41	0.77 → 0.61 (646)
OLMo 3.1 32B, zero-shot	1.27 → 1.33	0.37 → 0.25	0.47 → 0.45	0.53 / 0.70	0.70 / 0.79	0.73 → 0.67 (544)
Qwen3-8B, examples	1.29 → 1.34	0.37 → 0.15	0.54 → 0.40	0.55 / 0.68	0.74 / 0.63	0.66 → 0.53 (136)
Qwen2.5-7B, examples	1.16 → 1.33	0.38 → 0.21	0.55 → 0.41	0.55 / 0.62	0.65 / 0.53	0.73 → 0.60 (320)
Qwen2.5-32B, examples	1.20 → 1.43	0.12 → 0.10	0.42 → 0.38	0.79 / 0.89	0.81 / 0.78	0.84 → 0.55 (228)
OLMo 3 7B, examples	1.26 → 0.90	0.43 → 0.38	0.51 → 0.50	0.41 / 0.55	0.51 / 0.47	0.81 → 0.58 (648)
OLMo 3.1 32B, examples	1.12 → 1.37	0.27 → 0.20	0.48 → 0.42	0.67 / 0.83	0.77 / 0.76	0.72 → 0.62 (351)
Mistral-7B, examples	0.93 → 0.75	0.38 → 0.25	0.49 → 0.49	0.56 / 0.70	0.76 / 0.65	0.75 → 0.67 (236)

Table 34: Further checks (DDXPlus, T=0.7, base → post-trained). *Entropy*: of one list's renormalized probabilities, per case mean. *Lists that miss*: share of one model's lists that do not contain a severe true pathology among their five items (cases whose true pathology has severity 1–2), and that name none of the severe conditions the reference weights at least 0.1 (cases with at least one). *Accuracy at 50% coverage*: accuracy of the single-call answer on the half of cases ranked most confident by the stated probability or by 31-sample agreement. *Agreement AUROC*: on the cases where neither model's valid samples are unanimous, so both models are compared on the same cases and agreement is not tied.

+0.52; pneumonia and pulmonary embolism for Qwen2.5-32B-Instruct; unstable angina, panic attack and NSTEMI/STEMI for Qwen3-8B), but atrial fibrillation and influenza for OLMo-3.1-32B-Instruct, whose lists name acute pulmonary edema 0.25 less often.

F.7. List concentration, dangerous misses, selective prediction and common cases

Table 34 adds four checks computed with the same loader as the headline tables. *Concentration.* A five-item list could not get shorter, but its probabilities could concentrate. Zero-shot, the entropy of one list changes by −0.13 to +0.08 nats for the Qwen models and OLMo 3.1 32B, and with worked examples it rises for them (+0.05 to +0.25); only OLMo-3-7B-Instruct (−0.52 zero-shot) and Mistral-Instruct (−0.18 with examples) state markedly more concentrated lists. *Dangerous misses.* Per case, post-trained lists leave out a severe true pathology 9–18 points less often zero-shot and name none of the warranted severe conditions about 10 points less often (OLMo 3.1 32B −2.2, n.s.). *Selective prediction.* Ranking cases by agreement gives the higher accuracy at 50% coverage for every base model. After post-training the stated probability does better for Qwen3-8B and OLMo-3-7B-Instruct zero-shot and for every post-trained model with worked examples; the area under the risk–coverage curve gives the same ordering. *Common cases.* On the cases where both models give at least 16 valid samples, pass@16 over valid samples falls by 3.4–30.5 points zero-shot (n.s. for Qwen2.5-32B) and distinct answers by 1.5–2.7, so the collapse does not come from invalid answers; and on the cases where neither model's samples are unanimous, agreement's AUROC falls in every zero-shot pair (0.72–0.77 → 0.61–0.67).

F.8. A third family: Mistral-7B v0.3

We repeat the comparison with Mistral-7B-v0.3 and Mistral-7B-Instruct-v0.3, a family also studied by Jang et al. (2026). With the zero-shot prompt the base model does not answer: 52% of its samples at T=0.7 and 76% at T=1 are invalid (it continues the document instead), and its valid answers mostly repeat the first listed condition, so its single-call accuracy is 6% and the zero-shot pair is not informative. With the few-shot prompt of §F.1 the base model answers validly (0.4% invalid), and the pair reproduces the other families (Table 36). Post-training raises pass@1 by 14.0 points and lowers pass@32 by 14.5, cuts distinct answers from 5.0 to 1.3 and Cov@32 by 0.138, and moves the samples away from the reference (TV +0.058). Voting gains the base model 5.6 points and the Instruct model 0.8; the three most frequent of 32 samples beat one list by 12.6 points for the base model, while for the Instruct model the list is ahead by 7.0; agreement is the better error signal for the base model (AUROC 0.826 against 0.671) and the stated probability for the Instruct model (0.747 against 0.613). The stated list improves modestly (TV −0.036 and DDF1 +0.046, surviving clustering and Holm correction over 42 tests; −0.023 with the true pathology removed, which does not), the smallest significant gain of the four families, and 11 of the 17 severe conditions are named less often (3 significantly). Mistral-7B-Instruct-v0.3 also names 0.5 conditions per list that are not among the 49.

F.9. A medical-domain pair: MedGemma-4B

We add a medical-domain pair, MedGemma-4B's pretrained (`medgemma-4b-pt`) and instruction-tuned (`medgemma-4b-it`) releases, built on Gemma 3 4B and trained further on medical data, after the headline tests were fixed (DDXPlus, T=0.7, text only; the pretrained model uses the instruction-tuned model's chat template). Like Mistral's, its base model does not answer the zero-shot prompt (90% invalid samples), so the informative comparison gives both models the worked examples of §F.1 (Table 35; Holm over the 14 headline metrics, recall@5 over the 5 secondary ones). The samples collapse: distinct answers fall from 7.05 to 1.45, Cov@32 by 0.182 and pass@32 by 15.0 points [7.1, 23.6], and the samples move away from the reference (TV +0.075). But the base model's samples are so spread out that they reach 0.13 against another patient's diagnosis, and after subtracting that, post-trained minus base is −3.7 points (n.s.; Table 32); at each model's best prompt (the instruction-tuned model's is zero-shot) it is −5.4 raw and +5.6 corrected, both n.s. One list improves most in recall, reference mass and severe conditions (conditional TV is unchanged, +0.003): recall@5 +27.1 points, TV −0.077, reference mass +13.3 and warranted severe conditions named +15.0 (4 of 17 named less often). The error signals shift as for the Qwen models, more cleanly: voting gains the base model 10.7 points and the instruction-tuned model 0.8, agreement's AUROC falls from 0.878 to 0.677, and the stated probability rises from 0.657 to 0.813, overtaking agreement.

Appendix G. Thinking mode, aggregation and the error signal

G.1. Thinking mode

Qwen3-8B can reason in a thinking block before it answers; the main runs turn this off. With thinking on (zero-shot prompt, T=0.7, up to 8,192 tokens; traces average about

MedGemma-4B	Invalid	Sampled top answers pass@1	pass@32	Distinct	Cov@32	TV↓	One stated list Recall@5	TV↓	Severe	Vote@32 gain	AUROC Agree@32	Stated p
pretrained, zero-shot	90.0%	0.006	0.134	2.47	0.080	0.930	0.127	0.925	0.282	+0.4	0.607	0.518
instruction-tuned, zero-shot	1.3%	0.582	0.731	1.64	0.261	0.771	0.813	0.700	0.434	+0.0	0.707	0.675
pretrained, worked examples	5.5%	0.330	**0.785**	**7.05**	**0.412**	**0.712**	0.485	0.792	0.256	+10.7	**0.878**	0.657
instruction-tuned, worked examples	1.7%	**0.547**	0.635	1.45	0.230	0.787	**0.756**	**0.716**	**0.406**	+0.8	0.677	**0.813**

Table 35: Medical-domain pair (DDXPlus, T=0.7, 980 cases, 32 samples), added after the headline tests were fixed. The pretrained model's zero-shot answers are mostly invalid, so the worked-example rows are the comparison. Severe: share of warranted severe conditions one list names. Bold: better model with worked examples; every bold difference survives Holm correction over the pair's 14 headline metrics (recall@5: over the 5 secondary ones; pass@32 after chance correction does not survive: −3.7, n.s.).

1,500 tokens and all end before the limit), the same weights behave differently (Table 36). Accuracy rises at both ends: pass@1 is 0.671 against 0.559 without thinking, and pass@32 is 0.815, above the non-thinking model (0.607) and level with the base model (0.732 zero-shot, +8.4 [1.7, 15.4] points, not surviving Holm correction over 42 tests; 0.809 with worked examples). The samples are still narrow (1.76 distinct answers against 3.77 for the base model; Cov@32 0.283 against 0.323) but no farther from the reference than the base model's (TV 0.745 against 0.750). The stated list is the closest of any 7–8B model (TV 0.670, against 0.700 without thinking). The shift in error signals reverses: agreement across 31 samples is again the better signal (AUROC 0.732 against 0.638; difference 0.094 [0.027, 0.170]), as for base models, and voting gains 1.5 points (n.s.). The narrowing of the samples and the advantage of the stated probability are therefore properties of answering directly, which a reasoning mode largely undoes, consistent with the chain-of-thought prompt of §D.3.

G.2. Averaging stated lists across calls

Stated lists can be averaged across calls, as samples can be voted. For each case we average the first 5 or 32 valid stated lists, each renormalized over the conditions it names, and score the mean against the reference (Table 37). Averaging helps base models far more than post-trained ones, whose lists vary little between calls (at T=0.7 with the zero-shot prompt, 32 averaged lists name 7.6–13.4 conditions, against 16.6–22.0 for base models). The mean of five base-model lists is about as close to the reference as one post-trained list (Qwen3-8B 0.699 against 0.700, Qwen2.5-7B 0.717 against 0.717, OLMo 3.1 32B 0.691 against 0.701). With five lists on both sides, the post-trained mean is closer for Qwen2.5-7B under both prompts (−0.025 and −0.027) and for Qwen2.5-32B and Qwen3-8B under one prompt each, not significantly closer for OLMo 3.1 32B (zero-shot) and Mistral (few-shot), and farther for OLMo 3 7B and few-shot OLMo 3.1 32B. Averaged base lists also name severe conditions at least as often, and more often for OLMo 3.1 32B, OLMo 3 7B, Mistral and few-shot Qwen2.5-32B (by 0.06–0.14). One post-trained call thus gets about as close to the reference as several base calls; the packing test (Table 38) shows that it is a stabilized version of the base model's view of the patient rather than its literal average, and, as for the open-ended questions of Lake et al. (2025), post-training adds little that averaged base lists lack (five

Model	Invalid (%)	pass@1	pass@32	Distinct	Cov@32	TV, one list all	TV, one list w/o true	DDF1	TV samples	Vote@32 gain	Top-3 list − samples	AUROC stated	AUROC agree@5
Mistral-7B-v0.3 (base), few-shot	0.4	0.428	**0.791**	**5.02**	**0.355**	0.777	0.875	0.250	**0.741**	+5.6	−12.6	0.671	0.826
Mistral-7B-Instruct-v0.3, few-shot	0.3	**0.568**	0.646	1.34	0.217	**0.741**	**0.851**	**0.296**	0.799	+0.8	+7.0	0.747	0.613
Qwen3-8B-Base	11.3	0.388	0.732	**3.77**	**0.323**	0.743	0.833	0.276	0.750	+3.4	−7.3	0.595	0.772
Qwen3-8B, non-thinking	0.0	0.559	0.607	1.27	0.217	0.700	0.815	0.318	0.793	+0.3	+13.7	0.734	0.613
Qwen3-8B, thinking	0.0	**0.671**	**0.815**	1.76	0.283	**0.670**	**0.799**	**0.349**	**0.745**	+1.5	+3.0	0.638	0.693

Table 36: A third family and a reasoning mode on DDXPlus (T=0.7, 980 cases, 32 samples). Top: Mistral-7B v0.3 with the few-shot prompt of §F.1. Bottom: Qwen3-8B-Base and Qwen3-8B with thinking off (the main runs) and on, with the zero-shot prompt. Columns as in Table 23; Distinct: distinct top answers per 32 samples. Bold: best within each block.

Pair	TV, base model 1 list	TV, base model mean of 5	TV, base model mean of 32	TV, post-trained model 1 list	TV, post-trained model mean of 5	TV, post-trained model mean of 32	Post − base, 5 lists each TV	Post − base, 5 lists each Severe named
Zero-shot prompt								
Qwen3-8B	0.743	0.699	0.691	0.700	0.695	0.692	−0.004†	+0.030†
Qwen2.5-7B	0.779	0.717	0.706	0.717	0.692	0.685	−0.025	+0.023†
Qwen2.5-32B	0.726	0.681	0.674	0.674	0.658	0.654	−0.024	−0.037†
OLMo 3 7B	0.794	0.730	0.717	0.823	0.805	0.801	+0.074	−0.076
OLMo 3.1 32B	0.746	0.691	0.678	0.701	0.677	0.673	−0.014†	−0.088
Few-shot prompt								
Qwen3-8B	0.768	0.704	0.689	0.682	0.673	0.672	−0.030	−0.006†
Qwen2.5-7B	0.784	0.712	0.694	0.702	0.686	0.683	−0.027	+0.012†
Qwen2.5-32B	0.712	0.655	0.641	0.654	0.640	0.638	−0.015†	−0.094
OLMo 3 7B	0.787	0.703	0.676	0.808	0.786	0.778	+0.083	−0.141
OLMo 3.1 32B	0.719	0.651	0.630	0.697	0.682	0.680	+0.031	−0.060
Mistral-7B v0.3	0.777	0.708	0.691	0.741	0.723	0.720	+0.015†	−0.118
Zero-shot prompt, thinking on								
Qwen3-8B, thinking	0.743	0.699	0.691	0.670	0.650	0.647	−0.049	+0.129

Table 37: Averaging stated lists across calls (DDXPlus, T=0.7, 980 cases). TV to the reference differential of one stated list (per-case mean over valid lists, as in Table 13) and of the mean of the first 5 or 32 valid lists, for the base and the final post-trained model. Last two columns: post-trained minus base with five averaged lists each, for TV and for the share of severe conditions with reference probability ≥ 0.1 that are named. †: the condition-clustered 95% interval includes zero.

post-trained lists beat five base lists in four of eleven settings). Qwen3-8B's thinking mode goes furthest: five of its lists beat five base-model lists by more than any direct-answer model's (TV −0.049) and name more severe conditions (+0.129). For a post-trained model answering directly, averaging five lists instead of reading one improves TV by only 0.005–0.025 at T=0.7 (up to 0.038 at T=1); it names 6–17 points more severe conditions, mostly by naming 1.4 to 3.6 more conditions. Five post-trained lists still cover 0.44–0.50 of the reference mass at T=0.7 (OLMo 3 7B 0.37) with six to eight conditions, about twice what their 32 samples cover (0.22–0.28) and as much as a uniform guesser naming about 24 conditions (0.47), so the coverage deficit of §4.1 is one of the sampled readout.

Pair	TV of a post-trained list to own base mean	TV of a post-trained list to same-diagnosis means	reference	Within-model benchmark base list	Within-model benchmark post list	Differences own − same dx	Differences own − base list
Zero-shot prompt							
Qwen3-8B, T=0.7	0.368	0.444	0.701	0.372	0.109	−0.076	−0.004†
Qwen3-8B, T=1	0.424	0.486	0.701	0.551	0.153	−0.062	−0.128
Qwen2.5-7B, T=0.7	0.464	0.519	0.717	0.470	0.248	−0.055	−0.006†
Qwen2.5-7B, T=1	0.532	0.575	0.717	0.650	0.315	−0.043	−0.119
Qwen2.5-32B, T=0.7	0.360	0.431	0.674	0.375	0.189	−0.071	−0.015†
OLMo 3 7B, T=0.7	0.628	0.658	0.823	0.493	0.238	−0.031	+0.135
OLMo 3 7B, T=1	0.651	0.679	0.818	0.691	0.326	−0.028	−0.040
OLMo 3.1 32B, T=0.7	0.386	0.452	0.700	0.432	0.227	−0.066	−0.046
Few-shot prompt							
Qwen3-8B	0.469	0.518	0.681	0.471	0.102	−0.049	−0.003†
Qwen2.5-7B	0.559	0.596	0.702	0.515	0.171	−0.037	+0.044
Qwen2.5-32B	0.465	0.511	0.654	0.441	0.154	−0.045	+0.024
OLMo 3 7B	0.669	0.710	0.808	0.594	0.259	−0.040	+0.075
OLMo 3.1 32B	0.482	0.532	0.697	0.469	0.150	−0.050	+0.014†
Mistral-7B	0.495	0.546	0.740	0.495	0.181	−0.051	−0.000†
Thinking on							
Qwen3-8B, thinking	0.412	0.468	0.669	0.372	0.189	−0.057	+0.040

Table 38: Does a post-trained list reproduce its base model's view of the patient? (DDXPlus, 980 cases.) TV between a post-trained list and the mean of the base model's valid lists for the same case (own base mean), the base model's means for the other cases with the same true pathology, and the reference. Within-model benchmarks: a base list against the mean of the base model's other lists for the case, and a post-trained list against the mean of the post-trained model's other lists. †: the condition-clustered 95% interval includes zero.

Model (T=0.7)	Stated p	Agree@5	Agree@32
Qwen3-8B-Base	+0.042	+0.005	+0.024
Qwen3-8B	+0.116	+0.056	+0.098
Qwen2.5-7B	+0.044	+0.025	+0.038
Qwen2.5-7B-Instruct	+0.070	+0.054	+0.070
Qwen2.5-32B	+0.064	−0.026	−0.038
Qwen2.5-32B-Instruct	−0.043	−0.060	−0.109
OLMo-3-1025-7B	+0.083	−0.048	−0.052
OLMo-3-7B-Instruct	−0.067	−0.073	−0.049
OLMo-3-1125-32B	−0.005	+0.020	+0.011
OLMo-3.1-32B-Instruct	+0.063	+0.014	−0.023

Table 39: Spearman correlation between each uncertainty signal (lower confidence or agreement) and the entropy of the case's reference differential (DDXPlus, 980 cases). Neither readout tracks how ambiguous a case is.

G.3. Which signal to use: combination and held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen3-8B (thinking)	0.672	+1.5	+3.0	0.638	0.693	**0.732**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
Qwen2.5-32B	0.510	+7.0	−8.3	0.730	0.759	**0.797**
Qwen2.5-32B-Instruct	0.598	+1.9	+5.6	0.709	0.663	**0.721**
OLMo-3 7B base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3-7B-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
OLMo-3.1 32B base	0.441	+8.4	−13.2	0.629	0.805	**0.831**
OLMo-3.1-32B-Instruct	0.562	+0.9	−0.6	0.696	0.712	**0.772**
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 40: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call accuracy: first valid sample. Vote@32 gain: majority answer over valid samples minus single call (points). Top-3: one list's top three minus the three most frequent sampled answers, containing the true pathology (points). AUROC for the single-call answer being correct; bold marks the best signal.

models (including three fine-tuned from Llama-3.1-8B, HuatuoGPT-o1-8B, AlphaMed-8B and UltraMedical-8B, which appear only here), our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. With five-sample agreement, as in the audit below, combining the signals beats choosing between them; with agreement across all 31 further samples the advantage shrinks (post-trained combinations: agreement 0.742, rank average 0.761, +0.019 $[-0.007, 0.040]$; logistic 0.772, +0.030 $[0.013, 0.044]$), and for the 17 base combinations agreement alone (0.821) beats the rank average (0.780) and ties the logistic combination (0.827). The rest of this section uses five-sample agreement. Fitting a logistic model of correctness on the stated probability and five-sample agreement with 100 labeled cases, and evaluating on the other 880, gives a mean held-out AUROC of 0.753 for the 39 post-trained combinations (0.765 for all 56); averaging the two signals' ranks without fitting gives 0.746; picking the better single signal on the same 100 cases gives 0.720; always using agreement 0.695; always using the stated probability 0.671. The logistic combination beats the single-signal pick for 35 of the 39

post-trained combinations. This matches the general finding that hybrids of consistency and verbalized confidence beat either (Xiong et al., 2024). For choosing a single signal, the simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus. Neither signal tracks how ambiguous a case is: their Spearman correlations with the entropy of the reference differential are at most 0.12 in absolute value (Table 39).

Appendix H. Real cases and medical post-training

H.1. Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed matches or beats its parent at every k (Figure 5): +3.9 points at k=1 ([−0.5, +8.5], n.s.), widening to +8.4 at k=32 ([+2.7, +14.8]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.28 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points, significant only for the mild tier; Table 8). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 41). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+8.1, +20.0] and 18.1 [+10.4, +26.2] points. m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.16 and 0.20 vs. 0.11). In these three models, medical post-training on an instruction-tuned model does not add to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper (Liu et al., 2025a) describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.105**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.109
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.164
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.203
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.289
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.181

Table 41: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the sampled differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best zero-shot pass@32 of any 7B model. Bold: best value over all six models.

checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 41, lower block). It does not. RL from the base model improves on the base at every k (+14.3 [+10.4, +18.5] points at k=1, +9.5 [+5.0, +15.0] at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.1 [−3.0, +7.3] points worse at k=1 (n.s.) and 10.8 [+5.6, +16.5] points better at k=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−5.7, +5.3] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 8; all well below the chance levels of §D.2). RL from a base model also narrows less in OLMo's RL-Zero at T=1 and in our GRPO runs (§I), but not uniformly: RL-Zero at T=0.7 loses about as much as SFT, and within OLMo 3 7B's instruct pipeline the RL stage after SFT and DPO is the largest single narrowing (§D.4).

H.2. Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 3.7–7.0% for the Qwen-family models, and lower for OLMo 3 7B (2.5–4.5%). Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. The RL-trained medical models (AlphaMed; HuatuoGPT-o1 after SFT) thus state much higher confidence where they are least accurate. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident Qwen-family model (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at $k \leq 16$ (+0.6 [+0.1, +1.1] points at k=1; +1.0 [+0.0, +2.0], p=0.06, at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 4$ (+1.4 [+0.3, +2.6] at k=32; intervals resample the 749 diseases). These gains are an order of magnitude smaller than

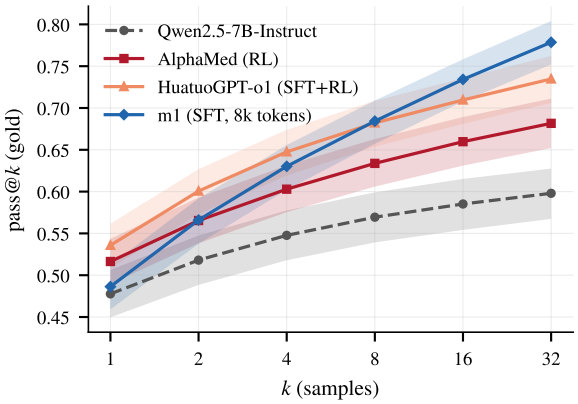


Figure 5: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: none of the three shows a sharpening tax at the medical stage.

the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 ($+0.7$ $[+0.3, +1.2]$) and trails it from k=8 on, by 2.2 $[1.2, 3.3]$ points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§I) are indistinguishable from their base (all pass@k differences within $\pm$0.6 points).

Accuracy depends strongly on when the source case report was published (Table 42). For every model in the table, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

H.3. Published case reports

MedCaseReasoning presentations stop before the authors' differential, so these cases test open-ended diagnosis from real presentations (prompts and grading in Appendix B). We sampled the five zero-shot Qwen and OLMo pairs (OLMo 3.1 32B's base model was added after the first analysis, with the same 4,096-token budget); Mistral is left out because its base

Model	≤2022 (n=901) pass@1	pass@32	2023–24 (n=348) pass@1	pass@32	2025+ (n=152) pass@1	pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 42: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

model needs worked examples, which this dataset cannot supply without reference lists. The two judges agree on 97.5% of 500,871 distinct names (κ 0.77; with only 5.0–7.0% of names accepted, most agreement is on rejections); Phi-4 accepts more names, and the strict grading leaves every direction unchanged. The authors' alternatives were extracted by a language model from the reports' reasoning notes; we drop the 148 of 3,168 that are only category words ("infection", "malignancy", "viral infection"), which leaves 886 cases with at least one and moves no coverage figure by more than 0.4 points. We checked the judges against silver labels from string matching. Both accept 99.9–100% of the sampled top answers whose words match the final diagnosis's (12,814), and at most 3.1% (Mistral-Small) and 5.2% (Phi-4) of those sharing no content word with it (189,788), an upper bound on false positives because abbreviations and synonyms fall among them. For the primary and strict grading these rates differ between base and post-trained models by at most 0.5 points (all intervals include zero); Phi-4 alone accepts 1.2 points fewer unrelated names from Qwen3-8B than from its base. The alternatives judge maps 94–97% of exact matches to the right item and 5–7% of names unrelated to every item to some item. The 17–34% of top answers between the two silver classes are not checked. For voting, agreement and distinct answers, top answers are grouped by their normalized text whether correct or not. Grouping paraphrases by Mondo concept instead (with the normalizer used for Phenopacket Store, which resolves 30% of distinct top-answer names; the rest keep their text) lowers distinct answers by 11–23% for every model, leaves every post-trained minus base difference in distinct answers significant and in the same direction, keeps voting gains at or below 3.9 points and raises agreement's AUROC by 0.01–0.04; after post-training the stated probability then remains significantly better than agreement only for Qwen3-8B (+0.063 $[0.001, 0.128]$) and OLMo 3.1 32B (+0.091 $[0.042, 0.143]$; Qwen2.5-32B +0.020, n.s.), and for Qwen3-8B-Base and OLMo-3-1125-32B agreement becomes significantly better (-0.077 $[-0.144, -0.009]$ and -0.073 $[-0.135, -0.009]$). Merging, on top of Mondo grouping, the names the judges matched to the final diagnosis or to the same authors' alternative lowers distinct answers by a further 12–27%; the Qwen pairs still lose distinct answers (-4.7, -1.8, -2.3), and the OLMo Instruct models' extra answers shrink from +8.7 to +6.0 and from +2.7 to +0.7 $[0.3, 1.1]$.

Unmatched paraphrases stay separate, so these counts still overstate the number of distinct diagnoses. To bound per-case judge false positives, which could favor base models because they give more distinct names, we also discarded every judge acceptance whose name shares no content word with its target (the final diagnosis, or the authors' item it was matched to). Every model's scores fall slightly and base models' a little more (coverage of the alternatives by 32 samples: Qwen2.5-7B base −4.9 points, Instruct −2.9), so the post-trained losses in that coverage shrink from 8–14 to 7–13 points for the Qwen and OLMo 3.1 32B pairs and keep their direction; Qwen2.5-32B's pass@32 loss grows to −3.7 (n.s. after Holm). Against the alternatives the authors wrote down, the stated/sampled split replicates (Table 43): post-trained samples cover 8–14 points fewer of them for the Qwen pairs and OLMo 3.1 32B (−9.4; −2.9, n.s., for OLMo 3 7B), one list covers them as before (equivalent within 5 points in every pair, TOST at the 5% level), and for Qwen3-8B, Qwen2.5-32B and OLMo 3.1 32B one list covers more of them than 32 samples, whereas base models' samples cover more than their lists (by 1.1–10.2 points, n.s. for Qwen2.5-32B). For the final diagnosis the reversal does not hold: 32 samples contain it more often than one list for every base model and every post-trained model but Qwen3-8B (0.308 against its list's 0.352). The consequences follow DDXPlus in direction but are smaller. Voting helps little for any model (+1.6 to +3.8 points for base models, −0.7 to +3.0 after post-training), partly because paraphrases of the same diagnosis count as different answers. After post-training the stated probability ranks correctness at least as well as agreement for Qwen3-8B (+0.085 $[0.022, 0.148]$), Qwen2.5-32B (+0.062 $[0.006, 0.117]$) and OLMo 3.1 32B (+0.114 $[0.064, 0.166]$), all non-inferior at a margin of 0.03 AUROC; for the other two the difference is not significant and non-inferiority is not shown, and it gives higher accuracy at 50% coverage for all five post-trained models; for base models agreement is not significantly different or, for OLMo 3 7B, better (−0.172). OLMo-3-7B-Instruct gives more distinct answers than its base model (24.6 against 13.7 among valid samples), whose answers are valid only 67% of the time, and so does OLMo-3.1-32B-Instruct (18.8 against 13.3; base valid 80%). Accuracy does not differ between cases published up to 2022 and from 2023 (e.g. Qwen2.5-32B-Instruct pass@32 0.49 and 0.52); since both periods may precede the models' training cutoffs, this does not rule out memorization.

Sets of equal size. Base-model samples cover the most alternatives (0.33–0.38), partly because 32 of them name more conditions. Comparing a list's first m items with the m most frequent sampled answers ($m = \min(3, \text{distinct answers})$; case bootstrap), one list covers more of the authors' alternatives than the samples within every post-trained model (+2.7 to +6.9 points) and within the Qwen and OLMo 3.1 32B base models (+1.2, n.s., to +5.0; OLMo 3 7B base −1.6). Against the base model's three most frequent answers, a post-trained list's top three cover as many or more for the Qwen pairs (+2.9 $[1.4, 4.4]$, +0.4 $[-1.1, 1.7]$, +4.6 $[3.3, 6.0]$) and OLMo 3.1 32B (+2.3 $[0.8, 3.8]$), and fewer for OLMo 3 7B (−2.5 $[-4.1, -1.1]$).

H.4. Closed-set case reports

To check the case-report results without a judge model, we gave each presentation a list of ten candidates: the final diagnosis, the authors' alternatives (category-only entries removed, at most nine per case; 2,935 in all) and, to fill the list, final diagnoses of other cases that

Pair	Model	Valid	pass@1	pass@32	Distinct	Recall@5 one list	Authors' alternatives one list	Authors' alternatives 32 samples	Vote@32 − single	AUROC, single-call answer correct stated p	AUROC, single-call answer correct agree@32	AUROC, single-call answer correct stated − agree
Qwen3-8B	base	0.878	0.139	0.433	13.02	0.301	0.300	0.330	+1.8	0.558	0.602	−0.044†
	post-trained	1.000	0.180	0.308	6.63	0.352	0.296	0.190	+0.8	0.678	0.592	+0.085
Qwen2.5-7B	base	0.986	0.134	0.438	15.75	0.263	0.286	0.376	+1.6	0.565	0.594	−0.029†
	post-trained	1.000	0.154	0.407	13.73	0.293	0.281	0.288	+0.8	0.623	0.593	+0.030†
Qwen2.5-32B	base	0.985	0.206	0.530	13.77	0.371	0.319	0.331	+2.7	0.666	0.621	+0.045†
	post-trained	1.000	0.251	0.495	11.51	0.429	0.318	0.247	−0.7	0.650	0.588	+0.062
OLMo 3 7B	base	0.669	0.073	0.390	13.70	0.221	0.248	0.351	+2.2	0.462	0.635	−0.172
	post-trained	0.994	0.120	0.370	24.61	0.240	0.236	0.322	+3.0	0.675	0.653	+0.023†
OLMo 3.1 32B	base	0.799	0.149	0.536	13.30	0.343	0.293	0.359	+3.8	0.582	0.631	−0.048†
	post-trained	1.000	0.224	0.499	18.76	0.419	0.294	0.266	+0.6	0.697	0.583	+0.114

Pair	Post-trained − base (primary grading) pass@1	pass@32	Distinct	recall@5	alt., list	alt., samples	List − samples (post, alt.)	Vote − single (post)	Strict grading pass@32	Strict grading recall@5
Qwen3-8B	+4.1	−12.5	−6.39	+5.1	−0.4†	−14.1	+10.6	+0.8†	−12.2	+5.0
Qwen2.5-7B	+2.0	−3.1†	−2.02	+3.0	−0.5†	−8.8	−0.7†	+0.8†	−3.2†	+2.8
Qwen2.5-32B	+4.4	−3.5†	−2.27	+5.8	−0.1†	−8.3	+7.1	−0.7†	−3.0†	+5.7
OLMo 3 7B	+4.7	−2.0†	+10.91	+1.9†	−1.3†	−2.9†	−8.6	+3.0	−1.6†	+1.8†
OLMo 3.1 32B	+7.5	−3.7†	+5.46	+7.6	+0.1†	−9.4	+2.9	+0.6†	−3.5†	+7.4

Table 43: Published case reports (MedCaseReasoning test split, 897 PMC case reports, zero-shot, 32 samples, T=0.7). A name is correct if the primary judge (Mistral-Small-24B-Instruct) says it names the report's final diagnosis; *strict*: only if Phi-4 agrees as well (the judges agree on 97.5% of 500,871 judged names, Cohen's κ 0.77). *Authors' alternatives*: share of the diagnoses other than the final one that the report's authors considered (extracted from its reasoning notes, without entries that are only category words such as "infection"; 886 cases with at least one, 3.4 on average) that one list names, or that any of the 32 sampled top answers names; *list − samples*: the difference within the post-trained model. Invalid samples count as wrong for pass@k; the list and agreement columns use valid samples. Distinct: distinct top answers among valid samples, grouped by normalized text whether correct or not (as for voting and agreement). Differences in points. †: not significant (paired case bootstrap, 10,000 resamples, Holm over the 45 pair tests; for the AUROC and list − samples differences, the 95% interval includes zero).

share no content word with any listed candidate, in random order. Answers are matched to the candidates by name (Table 44). A patient-blind uniform guesser would name 0.97 of the alternatives with 32 samples and 0.50 with one five-item list; every model's lists lie well above that (0.75–0.88) and every model's samples well below it (0.24–0.64), so this check measures the change in breadth between models that both concentrate, not distance from noise. The split replicates in all four pairs, more sharply than with open-ended answers: after post-training the 32 samples cover 22–27 points fewer of the authors' alternatives and pass@32 falls by 11.5–23.6 points, while one list contains the final diagnosis 4.1–7.5 points more often and covers 2.5–6.9 points more of the alternatives. Within each post-trained model one list covers 37–64 points more of the alternatives than all 32 samples together, against 11–34 points for base models. With ten candidates a five-item list names half of them, so list coverage is partly mechanical; the contrasts between models and between readouts are not. Agreement's AUROC falls in all four pairs; after post-training the stated probability is the better signal for Qwen3-8B (+0.104 [0.068, 0.140]), agreement for Qwen2.5-7B-Instruct

Pair	Model	Valid	pass@1	pass@32	Distinct	Recall@5 one list	Authors' alternatives one list	Authors' alternatives 32 samples	AUROC, single-call answer stated p	AUROC, single-call answer agree@32
Qwen3-8B	base	0.883	0.406	0.763	2.29	0.893	0.841	0.506	0.567	0.641
	post-trained	0.997	0.489	0.526	1.12	0.940	0.871	0.235	0.648	0.544
Qwen2.5-7B	base	0.966	0.420	0.779	2.76	0.836	0.765	0.555	0.513	0.676
	post-trained	0.992	0.470	0.580	1.40	0.911	0.833	0.287	0.543	0.590
Qwen2.5-32B	base	0.971	0.506	0.831	2.45	0.916	0.857	0.520	0.639	0.711
	post-trained	0.999	0.576	0.716	1.47	0.957	0.882	0.282	0.635	0.654
OLMo 3 7B	base	0.549	0.178	0.711	3.23	0.765	0.754	0.639	0.542	0.637
	post-trained	0.954	0.332	0.545	1.85	0.831	0.790	0.416	0.597	0.567

Pair	Post-trained − base pass@1	pass@32	Distinct	recall@5	alt., list	alt., samples	Alternatives: list − samples post	base	Vote@32 gain, post − base
Qwen3-8B	+8.4	−23.6	−1.17	+4.7	+3.0	−27.2	+63.6	+33.4	−0.8†
Qwen2.5-7B	+5.0	−20.0	−1.36	+7.5	+6.9	−26.8	+54.6	+21.0	−1.8†
Qwen2.5-32B	+7.0	−11.5	−0.98	+4.1	+2.5	−23.8	+60.0	+33.7	−2.7†
OLMo 3 7B	+15.4	−16.6	−1.38	+6.6	+3.6	−22.3	+37.4	+11.5	−3.0†

Table 44: Closed-set case reports (MedCaseReasoning, 897 cases, zero-shot, 32 samples, T=0.7). Each prompt lists ten candidates: the final diagnosis, the authors' alternatives (category-only entries removed) and, to fill the list, other cases' final diagnoses that share no content word with the candidates, in random order. Answers are matched to the candidates by name, with no judge; a sample is valid if its top answer is a candidate. The OLMo base model was capped at 1,024 tokens, and text after an invented chat turn is ignored. Differences in points; †: not significant (paired case bootstrap, 10,000 resamples, Holm across the 36 pair tests). Every list-minus-samples interval excludes zero.

(−0.047 [−0.090, −0.002]), and the two tie for the others. Voting changes accuracy by at most 4 points for any model.

H.5. Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.3, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 45). The DDXPlus shifts reappear, but the top-three comparison does not flip as it does on DDXPlus. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.07 for the post-trained ones, significantly only for Qwen3-8B (−0.056 [−0.120, −0.002]). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B (+0.4 [−0.8, +1.8]). The

Model	Single-call correct	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct			Correct answers
				Stated p	Agree@5	Agree@32	
Qwen3-8B-Base	0.074	+4.6	−7.0	0.567	0.768	**0.830**	104
Qwen3-8B	0.080	+0.4	+0.4	0.707	0.726	**0.763**	112
Qwen2.5-7B	0.062	+3.8	−5.4	0.598	0.788	**0.845**	87
Qwen2.5-7B-Instruct	0.056	+1.5	−3.6	0.717	0.661	**0.745**	78
OLMo-3-1025-7B	0.060	+3.1	−4.6	0.464	0.765	**0.814**	84
SFT	0.035	+1.1	−3.6	0.628	0.730	**0.814**	49
SFT+DPO	0.026	+1.2	−3.3	0.651	0.673	**0.725**	36
SFT+DPO+RLVR (Instruct)	0.035	+0.9	−2.7	0.652	0.644	**0.726**	49
RL-Zero (from base)	0.065	+2.3	−3.1	0.583	0.791	**0.827**	91
AlphaMed-7B	0.071	+1.6	−3.0	0.586	0.755	**0.797**	100
HuatuoGPT-o1-7B	0.093	+2.6	−5.7	0.653	0.708	**0.756**	130
m1-7B	0.084	+2.9	−7.9	0.650	0.747	**0.800**	117
AlphaMed-7B (base-RL)	0.054	+2.5	−4.2	0.672	0.733	**0.789**	75

Table 45: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, T=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 40; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

medical models behave like their post-trained parent, with agreement the better signal for all of them.

Appendix I. GRPO with a group-level reward

We also fine-tuned Qwen3-1.7B (instruct) and Qwen3-8B-Base with GRPO (LoRA, 200 steps, 16 samples per prompt, KL coefficient 0.04) on DDXPlus training patients disjoint from the test split, comparing the standard correctness reward (`acc`) with a group-level reward (`gm`). If k_a of the n samples in a GRPO group give top answer a, `gm` adds $\mathrm{clip}(n\,p_{\mathrm{ref}}(a) - (k_a - 1), 0, 1)$, which pays an answer only while its condition is under-represented relative to its reference mass. At 1.7B, a boosted variant (weight 3, training temperature 1.2, so the two changes are confounded) raised pass@1 by 1.6 and distinct answers by 0.10 over `acc` (two seeds; intervals ignore seed variation). From Qwen3-8B-Base, both recipes reached nearly the official Qwen3-8B's top-1 accuracy (0.537 and 0.539 vs. 0.559) while keeping more than twice its distinct answers (2.7–2.9 vs. 1.27), and `gm` narrowed the base model less than `acc` (distinct answers +0.14 at equal pass@1). Neither changed Phenopacket accuracy. Because `gm` needs a reference differential at training time, which real clinical data rarely provide, and the runs train on the evaluation distribution, these are existence results only. Closely related rewards computed from group answer counts have appeared since: a witness advantage that trains models to sample the distributions they state (Arabi et al., 2026), and a batch bonus that penalizes repeated answers (Song et al., 2025); we do not compare against them. Among the pooled analyses, the GRPO runs enter only the 56-combination audit (§G.3).

Pair (post − base)	pass@32 (pts)	Distinct	TV samples	TV one list	TV list w/o gold	DDF1	AUROC agree@32
Qwen3-8B, T=0.7	−12.4	−2.50	+0.043	−0.043	−0.018†	+0.042	−0.160
	[−19.6, −6.0]	[−2.89, −2.13]	[+0.021, +0.064]	[−0.059, −0.028]	[−0.032, −0.005]	[+0.029, +0.056]	[−0.216, −0.096]
Qwen3-8B, T=1	−14.6	−5.06	+0.067	−0.074	−0.039	+0.066	−0.158
	[−21.5, −8.3]	[−5.49, −4.65]	[+0.042, +0.091]	[−0.090, −0.059]	[−0.052, −0.027]	[+0.053, +0.080]	[−0.211, −0.104]
Qwen2.5-7B,	−9.7	−3.09	+0.028†	−0.062	−0.041	+0.062	−0.106
T=0.7	[−16.1, −4.1]	[−3.63, −2.57]	[+0.007, +0.049]	[−0.076, −0.049]	[−0.054, −0.028]	[+0.049, +0.075]	[−0.163, −0.044]
Qwen2.5-7B, T=1	−10.3	−5.59	+0.042†	−0.088	−0.056	+0.083	−0.104
	[−17.2, −4.2]	[−6.03, −5.15]	[+0.015, +0.068]	[−0.103, −0.074]	[−0.069, −0.043]	[+0.069, +0.097]	[−0.159, −0.046]
Qwen2.5-32B,	−4.9†	−2.10	+0.033	−0.053	−0.043	+0.051	−0.079
T=0.7	[−10.3, +0.5]	[−2.49, −1.74]	[+0.018, +0.049]	[−0.064, −0.042]	[−0.053, −0.032]	[+0.041, +0.061]	[−0.119, −0.038]
OLMo 3 7B,	−28.3	−2.56	+0.090	+0.029†	+0.034	−0.006†	−0.076†
T=0.7	[−35.6, −21.1]	[−2.84, −2.27]	[+0.071, +0.109]	[+0.004, +0.052]	[+0.017, +0.051]	[−0.022, +0.009]	[−0.157, +0.001]
OLMo 3 7B, T=1	−15.2	−3.95	+0.079	−0.009†	+0.006†	+0.024	−0.076†
	[−20.8, −9.7]	[−4.22, −3.68]	[+0.060, +0.098]	[−0.031, +0.011]	[−0.006, +0.018]	[+0.010, +0.036]	[−0.151, −0.011]
OLMo 3.1 32B,	−4.0†	−2.18	+0.020†	−0.045	−0.023	+0.040	−0.056†
T=0.7	[−6.8, −1.2]	[−2.47, −1.90]	[+0.002, +0.037]	[−0.059, −0.031]	[−0.034, −0.012]	[+0.026, +0.054]	[−0.095, −0.019]

Table 46: Base → post-trained (final Instruct) differences on DDXPlus with 95% intervals from a bootstrap over the 49 true-pathology clusters. †: not significant after Holm correction across 112 tests (50,000 resamples, add-one p-values). The AUROCs here take the single-call answer from the first sample whose list names any of the 49 conditions, as in all headline tests; Table 40 requires the top answer itself to be one of them, which changes the AUROC differences by at most 0.006.

Appendix J. Condition-clustered intervals and multiple comparisons

DDXPlus has 20 test cases for each of 49 true pathologies, and effects can be specific to a condition, so cases are not independent. Table 46 gives the headline base-vs-post differences with a bootstrap that resamples the 49 pathologies with all their cases (50,000 resamples for these headline tests, so that results near the threshold do not flip between runs; 10,000 elsewhere). Two-sided p-values use the add-one correction, $2(1 + \min(n_{\leq 0}, n_{\geq 0}))/(B + 1)$, so the smallest attainable value lies below the strictest Holm threshold, 0.05/112. Holm correction runs over the 112 tests (8 pairs × 14 metrics fixed in advance: pass@1, pass@32, distinct answers, coverage@32, sampled TV, list TV, mass named, severe named, list and sample TV without the true pathology, DDF1, unwarranted severe conditions named, and the two AUROCs; the table shows seven). The same procedure is applied separately to the few-shot pairs (70 tests), the Mistral and thinking-mode pairs (42) and the stage and medical comparisons (154). Not surviving correction among the headline tests shown, marked †: list TV without the true pathology for Qwen3-8B at T=0.7; sampled TV for Qwen2.5-7B at T=0.7; sampled TV for Qwen2.5-7B at T=1; pass@32 for Qwen2.5-32B at T=0.7; list TV for OLMo 3 7B at T=0.7; DDF1 for OLMo 3 7B at T=0.7; agreement AUROC for OLMo 3 7B at T=0.7; list TV for OLMo 3 7B at T=1; list TV without the true pathology for OLMo 3 7B at T=1; agreement AUROC for OLMo 3 7B at T=1; pass@32 for OLMo 3.1 32B at T=0.7; sampled TV for OLMo 3.1 32B at T=0.7; agreement AUROC for OLMo 3.1 32B at T=0.7. OLMo 3 7B's list TV without the true pathology at T=0.7 survives as a significant *increase*.

Pooled effects and equivalence. Statements such as "robust only for Qwen" compare significance across pairs. Table 47 instead pools each difference across pairs with a joint bootstrap over the 49 pathologies, which keeps the correlation between pairs scored on the same cases, and tests whether Qwen differs from the other families. The sampled

	Zero-shot (5 pairs)		Worked examples (6)		Another model's examples (6)	
Post-trained − base	pooled	Qwen − other	pooled	Qwen − other	pooled	Qwen − other
pass@32 (points)	−11.9 [−15.4, −8.5]	+7.1 [+2.6, +11.6]	−19.7 [−25.7, −14.1]	+2.9 [-2.8, +8.1]†	−18.1 [−23.8, −12.5]	+3.7 [-1.1, +8.2]†
distinct answers	−2.49 [−2.78, −2.20]	−0.20 [-0.53, +0.09]†	−3.86 [−4.47, −3.24]	+1.05 [+0.69, +1.41]	−3.86 [−4.44, −3.27]	+2.34 [+1.99, +2.71]
recall@5, one list (points)	+11.2 [+8.1, +14.4]	+6.8 [+3.6, +10.0]	+13.9 [+8.8, +18.9]	+4.9 [+0.1, +10.0]	+13.5 [+9.3, +17.8]	+3.6 [-1.1, +8.3]†
TV, one list	−0.035 [−0.045, −0.025]	−0.044 [−0.058, −0.031]	−0.044 [−0.058, −0.031]	−0.063 [−0.076, −0.050]	−0.048 [−0.060, −0.036]	−0.041 [−0.055, −0.029]
conditional TV	−0.031 [−0.043, −0.020]	−0.064 [−0.092, −0.037]	+0.004 [-0.018, +0.027]†	−0.052 [−0.078, −0.028]	−0.057 [−0.073, −0.040]	−0.034 [−0.055, −0.014]
severe named (points)	+7.2 [+4.3, +10.4]	+5.1 [+2.3, +7.8]	+5.4 [+1.6, +9.3]	+8.3 [+4.7, +11.7]	+4.9 [+1.7, +8.3]	+7.0 [+3.1, +11.0]
AUROC, agreement	−0.095 [−0.125, −0.061]	−0.049 [-0.098, +0.005]†	−0.190 [−0.229, −0.146]	−0.058 [−0.100, −0.010]	−0.182 [−0.225, −0.130]	−0.002 [-0.039, +0.037]†

Table 47: Pooled base → post-trained effects across pairs (DDXPlus, T=0.7). *Pooled*: mean of the pairs' differences; *Qwen − other*: mean over the Qwen pairs minus the mean over the OLMo (and Mistral) pairs. 95% intervals come from a joint bootstrap that draws the 49 pathologies once per resample and applies the draw to every pair, so the correlation between pairs scored on the same cases is kept (2,000 resamples). †: interval includes zero.

collapse, the stated-list gain in recall and the weaker agreement are significant when pooled in all three prompt settings; agreement's AUROC falls by 0.10 zero-shot and 0.18–0.19 with worked examples. The stated list gains more for Qwen than for the other families in TV, conditional TV and severe conditions named in every setting, and in recall zero-shot and with reference examples; the zero-shot pass@32 loss is smaller for Qwen (by 7.1 points), mostly because OLMo 3 7B loses 28; and agreement falls more for Qwen only with reference examples. For claims of no difference we use two one-sided tests with margins fixed in advance (5 points for accuracy and recall, 0.02 for TV, 0.03 for AUROC). Thinking mode's pass@32 is at least its base model's (90% interval $[+2.6, +14.1]$ points), but the non-significant zero-shot pass@32 losses at 32B and OLMo 3 7B's recall change (+1.5, $[-2.9, +5.9]$) are inconclusive rather than equivalent to zero.